Precision-Aware Hopfield Retrieval: Unifying Population Codes and Memory Retrieval with Information Optimization

Yi-Chun Hung[1] **Dennis Wu**[1,2] **Hong-Yu Chen**[1,2] **Han Liu**[1,2,3] **Emma Alexander**[1]

[1]Department of Computer Science, Northwestern University
[2]Center for Foundation Models and Generative AI (CFMG), Northwestern University
[3]Department of Statistics and Data Science, Northwestern University

ealexander@northwestern.edu | Project Website

Abstract

Bayesian estimation has long been used to study both continuous estimation and discrete associative memory, but how encoding precision and memory retrieval should be jointly optimized remains unclear. We introduce *precision-aware Hopfield retrieval*, a framework that uses the encoder's Fisher information to quantify encoding precision and incorporates it directly into the decoder of memory retrieval. Four results follow. First, optimal encoder-decoder information is shaped only by the prior, across L^p loss functions. Second, the optimal information follows a general coupling law across L^p loss functions. Third, optimal precision-aware Hopfield retrieval uses a stimulus-dependent inverse temperature. It outperforms the encoder-blind Bayesian decoder on both retrieval loss and required coding resource. Last, the framework yields a closed-form expression for memory recall bias. It reproduces central-tendency and cognitive-load effects observed in cognitive psychology. These effects emerge only when the encoder and decoder are modeled jointly.

1 Introduction

Encoder-decoder is a common language in both the biological and machine learning (ML) fields. In ML, an encoder maps input data to a latent space, and a decoder transforms this representation into the target space. Analogously, in biology, a population of neurons fires in response to input stimuli, while downstream neural circuits decode this activity for serving different tasks. Both sides of this framework are supported by well-developed theories [11, 27], and recent work [14, 15] has begun to bridge them for continuous stimulus estimation. An analogous bridge for memory retrieval, where decoding is discrete associative recall, remains underexplored.

On the decoding side, Hopfield networks (HNs) [18, 19] and their modern variants [29, 30, 39] are the commonly used models of discrete associative memory, with rich theory on capacity, retrieval error, and dynamics [22, 31, 39]. For analytical tractability, however, these analyses abstract the encoder away: memory patterns are assumed uniformly distributed and/or queries are perturbed by Gaussian noise [44]. How the structured, stimulus-dependent precision of a realistic encoder reshapes Hopfield retrieval remains unaddressed.

On the encoding side, efficient coding theory predicts that neural populations allocate more coding resources to more probable stimuli [11], shaping population geometry [28]. Recent work [14] derives a closed-form link from this geometry to perceptual bias, but the derivation assumes the decoder performs *Bayesian continuous estimation*. In cognitive science, memory retrieval exhibits systematic

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biases, including central tendency, where recalled values are attracted toward frequently experienced stimuli [4, 24, 25], and cognitive-load effects, where recall biases increase under greater memory demands [3, 56]. Whether these phenomena can be explained by the interaction between neural encoding precision and discrete associative memory retrieval remains open.

To answer the questions above, we introduce *precision-aware Hopfield retrieval*. The encoder's Fisher information enters the retrieval pipeline directly. Modeling the encoder and decoder jointly yields four main results.

- **Power-law optimal allocation and recall-capacity trade-off (Props. 1 and 2).** We show that the optimal memory density $\rho^*(s)$ and retrieval precision $\gamma^*(s)$ across stimulus space follow a power law of a given prior $p(s)$. We also analyze the trade-off of recall error and memory capacity in terms of $\rho(s)$ and $\gamma(s)$.
- **General coupling of precision and memory allocation (Thm. 1).** The jointly optimal retrieval precision $\gamma^*(s)$ and memory density $\rho^*(s)$ satisfy $\gamma^*(s) \propto [\rho^*(s)]^2$. We call this stimulus-varying relationship between retrieval precision and memory allocation the *general coupling*, and prove that it holds across all L^p losses with $p > 0$.
- **Encoder-blind Bayesian decoding is not optimal (Sec. 5.2).** Precision-aware decoding specifies stimulus-varying inverse temperatures $\beta^*(s)$ and outperforms encoder-blind Bayesian decoding on both retrieval loss and required coding resource
- **Closed-form for memory recall bias (Prop. 3).** The framework yields a closed-form expression for recall bias. It is consistent with empirical findings in cognitive psychology, including central-tendency effects and cognitive-load-dependent biases.

2 Related work

Population coding. Population codes study how sensory neural populations optimally encode stimuli under resource constraints, such as firing rate [11, 40, 51, 52, 55], coding capacity [50, 53], or metabolic cost [23, 52]. These studies also implicitly assume an efficient decoder so that the decoder information loss is negligible. A variety of coding objectives have been studied, such as maximizing mutual information [11, 12, 35, 52, 53], entropy [9, 32], and minimizing general L_p loss [11, 35, 51]. However, mutual information and L_p loss are generally intractable, so recent work instead adopts FI as a proxy [11, 12, 35, 51, 52], approximating them by $\log(\text{FI})$ and $\text{FI}^{-p/2}$, respectively. Although objectives and resource constraints differ across studies, the closed-form optima share a common *power-law* allocation $\text{FI}(s) \propto p(s)^{2\alpha}$, where the exponent α is determined by the choice of objective and constraint [11, 13, 23, 51].

Recent work [14, 38, 54] has extended this power-law allocation with a Bayesian decoder for continuous estimation. This bridges optimal resource allocation to perceptual quantities, specifically discrimination thresholds and perceptual biases. For population codes that maximize mutual information, $\text{FI}(s) \propto p(s)^2$, which has been observed in [11, 13]. In this setting, the Cramer Rao bound would predict perceptual discrimination thresholds that scale as $D(s) \propto p(s)^{-1}$, which has also been observed [11, 13]. As for bias, Wei and Stocker [54] derive the perceptual bias under a family of Bayesian decoders with different loss functions, and show that—despite differing in direction (attraction vs. repulsion)—the biases obey the closed-form lawful relation $b(s) \propto \frac{d}{ds}[D^2(s)]$. Later works [37, 38] generalize the bias derivation beyond infomax coding, and Hahn and Wei [14] decompose the bias additively into a prior-attraction term and a likelihood-repulsion term under general L^p loss. Yet this encoder–decoder bridge has only been instantiated for continuous Bayesian estimation. Whether the same population geometry governs biases in *discrete memory recall* remains open.

Associative memory. Hopfield networks (HNs) [18, 19] and their modern variants [29, 30, 39] model discrete associative memory through an energy landscape whose local minima are the stored patterns. Memory retrieval is energy descent from a noisy query and is specified by an update rule. Different choices of energy and update rule have been analyzed from complementary angles—memory capacity [22, 39], retrieval error [39], and retrieval dynamics [29, 41].

More specifically, these aspects are discussed through different components of the update rule: input characteristics [5], synaptic weights and connection order [6, 8], and the readout mechanism [21, 41, 49]. However, these analyses typically assume tractable noise, such as binary pattern flipping

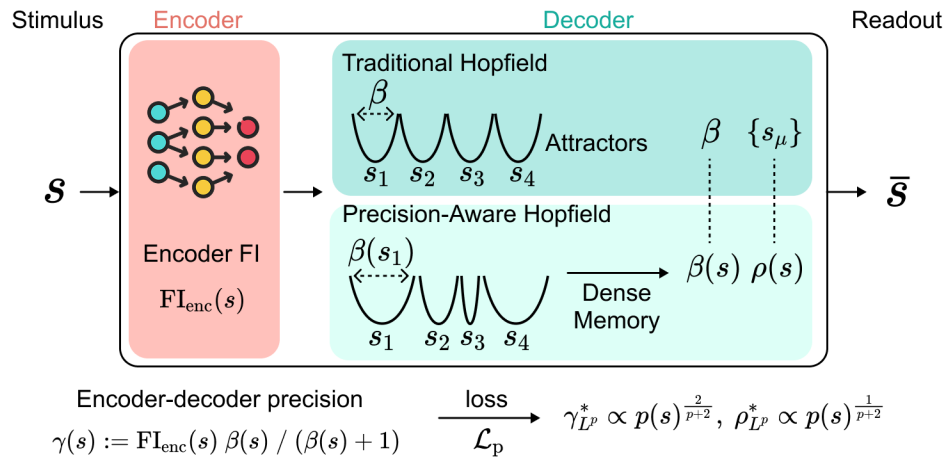


Figure 1: Overview of the precision-aware Hopfield framework. A stimulus s is encoded with Fisher information $\mathrm{FI}_{\mathrm{enc}}(s)$ and decoded by a Hopfield network (HN). Our framework (bottom) replaces an inverse temperature β and $\{s_\mu\}$ in traditional HNs (top) with stimulus-dependent $(\beta(s), \rho(s))$ in the dense memory regime (bottom). It produces an encoder-decoder precision $\gamma(s)$. Jointly optimizing under $\mathcal{L}_p$ satisfies the general coupling: $\gamma^*(s) \propto \rho^*(s)^2$ (Thm. 1).

[18, 30] or Gaussian perturbation [20, 36]. This may be an oversimplification, since noise is often data dependent [10, 26]. Recent work [1] takes a first step by forming memory patterns under the data manifold hypothesis, and could plausibly be extended to data-dependent noise. Even so, the input geometry is prescribed by a hand-designed generator rather than coupled to an encoder. As a result, how the stimulus-dependent precision of real encoders reshapes Hopfield retrieval is left unaddressed.

Bayesian encoder-decoder family. Our framework connects to the Bayesian encoder–decoder family through a tempered likelihood. Assuming a uniform prior over stored memories, the resulting posterior recovers the decoder in our framework (Eq. (2)). By allowing the inverse temperature $\beta(\mathbf{r})$ to vary, this formulation establishes connections between our framework and previous Bayesian models [33, 42, 43, 45].

When $\beta(\mathbf{r}) = 1$, the tempered likelihood reduces to the standard likelihood used in conventional Bayesian retrieval. For example, Sommer and Dayan [45] cast associative-memory retrieval as Bayesian inference by accounting for both cue corruption and storage errors in the synaptic memory matrix. Related work extended this probabilistic treatment to plasticity-matched retrieval and bounded metaplastic synapses [33, 43], while other studies examined context-dependent inference [17], familiarity- and age-dependent retrieval [42], and learned mechanisms for familiarity detection [47].

It is worth noting that the familiarity- and age-dependent retrieval of [42] can be interpreted as adapting an effective β according to the inferred age of a memory, whereas Savin et al. [47] explicitly introduced dynamic variation in β to facilitate retrieval from bounded metaplastic synapses. Together, these studies suggest that variations in β may play an important role in memory-retrieval dynamics and have potential biological relevance.

3 Precision-aware Hopfield retrieval

To bring encoder precision into Hopfield retrieval, we must define a new statistical decoding rule for the Hopfield model (Sec. 3.1), apply it in the dense memory regime (Sec. 3.2), and translate this to a new readout rule (Sec. 3.3).

3.1 Problem setup

We consider memory retrieval in two stages (Fig. 1). A scalar stimulus $s \in \mathcal{S}$ drives a population response $\mathbf{r} \in \mathbb{R}^N$ through an *encoder*. A *decoder* then retrieves one of M stored patterns $\{s_\mu\}_{\mu=1}^M$

from $\mathbf{r}$. Marginalizing over the population response gives the *per-pattern retrieval density*—the recall density per stored pattern,

$$p(s_\mu \mid s) = \int \underbrace{p(s_\mu \mid \mathbf{r})}_{\text{decoder}} \underbrace{p(\mathbf{r} \mid s)}_{\text{encoder}} \, d\mathbf{r}, \tag{1}$$

which fully specifies the input–output behavior of the system.

Encoder. The encoder is a population code with likelihood $p(\mathbf{r} \mid s)$. Its precision is summarized by the Fisher information $\mathrm{FI}_{\mathrm{enc}}(s)$. $\mathrm{FI}_{\mathrm{enc}}(s)$ absorbs the standard ingredients of efficient coding—gain, density, and dispersed noise factor [11, 23, 51].

Decoder. The decoder is a continuous Hopfield network whose retrieval follows the softmax rule with inverse temperature $\beta(\mathbf{r})$:

$$p(s_\mu \mid \mathbf{r}) = \frac{\exp\big(-\beta(\mathbf{r})\, \mathcal{E}(\mathbf{r}, s_\mu)\big)}{\sum_{\mu'} \exp\big(-\beta(\mathbf{r})\, \mathcal{E}(\mathbf{r}, s_{\mu'})\big)}, \tag{2}$$

where $\mathcal{E}(\mathbf{r}, s_\mu) := -\log p(\mathbf{r} \mid s_\mu)$ is the negative log-likelihood of the response. Setting $\beta(\mathbf{r})$ to a constant recovers the posterior mean (see Sec. G) and connects the model to attention mechanisms [39]. Allowing $\beta(\mathbf{r})$ to vary makes the decoder precision-aware via its dependence on $\mathbf{r}$. Under (A1) below, $\beta(\mathbf{r})$ reduces to a stimulus-dependent function $\beta(s)$ in the large-population limit, which is the form we use in the subsequent derivation.

3.2 Asymptotic closed form

The per-pattern retrieval density Eq. (1) is generally intractable. We derive a closed form under two assumptions:

- (A1) **Large population limit.** The number of neurons N is large.
- (A2) **Dense memory regime.** The number of stored patterns M is large enough that sums over memory indices can be replaced by integrals against a continuous memory density $\rho(s)$.

(A1) and the role of $\hat{s}$ in the decoder. (A1) implies two asymptotic-normality results. The maximum likelihood estimator $\hat{s} := \arg\max_s p(\mathbf{r} \mid s)$ becomes an asymptotically sufficient statistic for $\mathbf{r}$. Its sampling distribution satisfies $p(\hat{s} \mid s) \to \mathcal{N}\big(s,\, \mathrm{FI}_{\mathrm{enc}}(s)^{-1}\big)$. These two results pin down the form of $\beta(\mathbf{r})$. Asymptotic sufficiency lets us restrict $\beta(\mathbf{r}) = \tilde{\beta}(\hat{s}(\mathbf{r}))$ without loss of generality: any dependence on $\mathbf{r}$ beyond $\hat{s}$ is asymptotically redundant. Consistency, $\hat{s} \to s$, then gives $\tilde{\beta}(\hat{s}) \to \tilde{\beta}(s) =: \beta(s)$ in the large-population limit. The stimulus-dependent inverse temperature $\beta(s)$ thus emerges as an asymptotic object, not an a priori parameterization. The MLE $\hat{s}$ plays a dual role. It sets the local sharpness $\beta(\hat{s})$ as a noisy continuous estimate fed to the decoder. Memory retrieval then anchors this estimate to discrete prototypes $\{s_\mu\}$: the subsequent energy descent acts as a denoising process that snaps $\hat{s}$ to a stored pattern. As we will show, modeling this continuous-to-discrete step explicitly is required to reproduce cognitive biases such as central-tendency effects, which pure continuous estimation cannot fully capture.

(A2) and the discreteness of attractors. (A2) lets the partition function in the denominator of Eq. (2) be evaluated as a Gaussian integral. Crucially, evaluating the partition function via a continuous integral captures the macroscopic expected attention weights, rather than implying the merging of attractors into a continuous manifold. The exact discrete dynamics are preserved provided the retrieval precision maintains a sufficient signal-to-noise ratio ($\mathrm{SNR} \gg 1$); we provide an extended discussion of this physical regime in Sec. I.1.

Approximation sketch. A second-order Taylor expansion of $\mathcal{E}(\mathbf{r}, s_\mu)$ around $\hat{s}$ replaces the energy with $\frac{1}{2}\,\mathrm{FI}_{\mathrm{enc}}(s)\,(s_\mu - \hat{s})^2$ in the large-N limit, so the decoder becomes a Gaussian in s_μ centered at $\hat{s}$. Marginalizing $\hat{s}$ against its asymptotic Gaussian distribution yields a convolution of two Gaussians. The full derivation is given in Appendix B.

Approximation result. The per-pattern retrieval density is locally Gaussian:

$$p(s_\mu \mid s) \approx \frac{1}{\rho(s)} \mathcal{N}\big(s_\mu;\, s,\, \gamma(s)^{-1}\big)\,, \qquad \gamma(s) = \frac{\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)}{\beta(s)+1}. \tag{3}$$

Here, $1/\gamma(s) = 1/\mathrm{FI}_{\mathrm{enc}}(s) + 1/(\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s))$, so the total inverse precision decomposes additively into encoder and decoder contributions, consistent with independent noise channels. This is the information-theoretic analog of impedance matching. The encoder sets an irreducible ceiling, $\gamma(s) < \mathrm{FI}_{\mathrm{enc}}(s)$, regardless of β. The decoder only modulates how close the system approaches that ceiling. Exact Bayesian decoding, which uses a constant β, sacrifices stimulus-varying encoder precision. For example, when $\beta = 1$, Bayesian decoding recovers *only half of the encoder precision*. The consequences of this effect are analyzed in Prop. 2 and empirically demonstrated in Sec. 5.2.

Note that the encoder is parameterized by $\mathrm{FI}_{\mathrm{enc}}(s)$, while the decoder depends on $\beta(\mathbf{r})$ and the discrete patterns $\{s_\mu\}_{\mu=1}^{M}$. Under (A1) and (A2), we approximate these by a stimulus-dependent $\beta(s)$ and a continuous pattern density $\rho(s)$, respectively.

3.3 Hopfield readout

The per-pattern retrieval density in Eq. (3) normalizes against the memory measure: $\int p(s_\mu \mid s)\,\rho(s_\mu)\,ds_\mu = 1$. To extract a stimulus estimate from the network, we need a complementary object: a density over the output location $\bar{s}$. We define the *readout retrieval density* by weighting the per-pattern density with the local memory density:

$$p_{\mathrm{ret}}(\bar{s} \mid s) := \rho(\bar{s})\, p(s_\mu = \bar{s} \mid s) = \frac{\rho(\bar{s})}{\rho(s)} \mathcal{N}\big(\bar{s};\, s,\, \gamma(s)^{-1}\big)\,, \tag{4}$$

which is a proper density on stimulus space to leading order under (A2). The decoded stimulus estimate is the posterior mean of the retrieved location under p_{ret}:

$$\bar{s}_{\mathrm{out}}(s) := \mathbb{E}_{p_{\mathrm{ret}}}[\bar{s}]. \tag{5}$$

The shift of $\bar{s}_{\mathrm{out}}$ from s is the retrieval bias, derived and compared with literature in Sec. 5.

4 Optimal decoding and memory allocation

We introduce the retrieval objectives for L^p-loss (Sec. 4.1) and solve the optimal memory and precision allocation ($\rho^*(s)$ and $\gamma^*(s)$) under some constraints (Sec. 4.2). Then, we discover the general coupling in the optimal solutions (Sec. 4.3) and analyze the proposed framework (Sec. 4.4).

4.1 Retrieval objectives

We assess retrieval quality through the per-pattern retrieval density $p(s_\mu \mid s)$ from Eq. (3). L_p objectives are functionals of this density.

L^p loss. For $p > 0$, we define the pointwise L^p loss as the expected retrieval distortion weighted by the inverse local memory density,

$$\mathcal{L}_p(s) := \frac{1}{\rho(s)} \int |s_\mu - s|^p\, \rho(s_\mu)\, p(s_\mu \mid s)\, \mathrm{d}s_\mu \approx \frac{c_p}{\rho(s)}\, \gamma(s)^{-p/2}, \tag{6}$$

where $c_p = 2^{p/2}\,\Gamma\big(\frac{p+1}{2}\big)/\sqrt{\pi}$ is the p-th absolute moment of a standard Gaussian (see Sec. D for the derivation). The factor $1/\rho(s)$ introduces an explicit memory-density weighting, distinguishing this objective from the conventional expected retrieval distortion. Under the locally uniform memory density approximation, the loss can be approximated by directly integrating the retrieval distortion against the per-pattern retrieval probability.

Averaging over the stimulus prior yields the global objective

$$\mathcal{L}_p := \int p(s)\, \mathcal{L}_p(s)\, \mathrm{d}s = c_p \int p(s)\, \frac{\gamma(s)^{-p/2}}{\rho(s)}\, \mathrm{d}s. \tag{7}$$

We next jointly optimize the memory density $\rho(s)$ and retrieval precision $\gamma(s)$ under this objective, subject to resource constraints. For $p = 2$, the objective recovers the precision dependence of the classical discrimax criterion when memory density is fixed [11, 23].

Note that, unlike L^p losses, logarithmic loss, popular for its connection to mutual information [7, 50, 53], introduces a $\log \rho(s)$ term that favors reducing local memory density but removes the direct coupling to the $\log \gamma(s)$ term. Consequently, a shared resource constraint would be needed for joint optimization over memory density and retrieval precision for the log loss, which we leave to future work.

4.2 Constraints and joint optimization

We minimize the retrieval objective Eq. (7) subject to two budgets:

$$\int \rho(s)\,\mathrm{d}s = M, \qquad \int \sqrt{\gamma(s)}\,\mathrm{d}s = C. \tag{8}$$

The first caps the total number of stored patterns. The second bounds the total resolution the encoder–decoder system deploys across stimulus space. It is the natural analog of the efficient-coding budget $\int \sqrt{\mathrm{FI}_{\mathrm{enc}}(s)}\,\mathrm{d}s = \mathrm{const}$ from sensory neuroscience [51, 53]. The pointwise constraint $\gamma(s) \le \mathrm{FI}_{\mathrm{enc}}(s)$ inherited from Eq. (3) bounds C above by $C_{\max} := \int \sqrt{\mathrm{FI}_{\mathrm{enc}}(s)}\,\mathrm{d}s$. We further restrict C to a loss-dependent range $(0, C_{\mathrm{sat}})$ with $C_{\mathrm{sat}} \le C_{\max}$, chosen so that the interior optimum derived below satisfies $\gamma^*(s) < \mathrm{FI}_{\mathrm{enc}}(s)$ at every stimulus. This restriction keeps the optimal $\beta^*(s)$ finite everywhere and avoids the encoder-saturation boundary. The explicit threshold for L^p loss is derived in Sec. E.

Proposition 1 (Optimal memory and precision allocation)**.** For $p > 0$, fix $M, C > 0$ with $C \in (0, C_{\mathrm{sat}})$. The global retrieval objective Eq. (7) is minimized jointly over (ρ, γ), subject to the budgets Eq. (8), by the following closed-form allocations:

$$\rho^*(s) \propto p(s)^{\frac{1}{p+2}}, \qquad \gamma^*(s) \propto p(s)^{\frac{2}{p+2}}. \tag{9}$$

Proof. The full Lagrangian derivation is provided in Sec. E. □

Interpretation. The optimal memory density and retrieval precision follow power-law relationships with the stimulus prior. Both exponents decrease as p increases, reflecting the stronger penalty on large retrieval errors and driving the optimal allocations toward uniformity.

Optimal inverse temperature. Inverting the relation $\gamma(s) = \beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)/(\beta(s)+1)$ in Eq. (3) gives

$$\beta^*(s) = \frac{\gamma^*(s)}{\mathrm{FI}_{\mathrm{enc}}(s) - \gamma^*(s)}, \tag{10}$$

which remains finite at every s under the feasibility restriction $C < C_{\mathrm{sat}}$.

4.3 General coupling of optimal precision and memory allocation

The optimal allocations in Eq. (9) reveal a structural relationship between retrieval precision and memory density that holds across all L^p losses with $p > 0$.

Theorem 1 (General coupling)**.** *For any $p > 0$, fix $M, C > 0$ with $C \in (0, C_{\mathrm{sat}})$. The optimal allocation under the budgets Eq.* (8) *satisfies*

$$\gamma^*(s) = \kappa\,[\rho^*(s)]^2, \tag{11}$$

where $\kappa = (C/M)^2 > 0$ is a global constant independent of the stimulus and loss exponent p.

Proof. From Eq. (9), the optimal allocations satisfy $\rho^*(s) \propto p(s)^{1/(p+2)}$ and $\gamma^*(s) \propto p(s)^{2/(p+2)}$. Hence, $\gamma^*(s) \propto [\rho^*(s)]^2$ for any $p > 0$. Imposing the budgets $\int \rho^*(s)\,\mathrm{d}s = M$ and $\int \sqrt{\gamma^*(s)}\,\mathrm{d}s = C$ yields $\sqrt{\gamma^*(s)} = (C/M)\rho^*(s)$, establishing $\kappa = (C/M)^2$. □

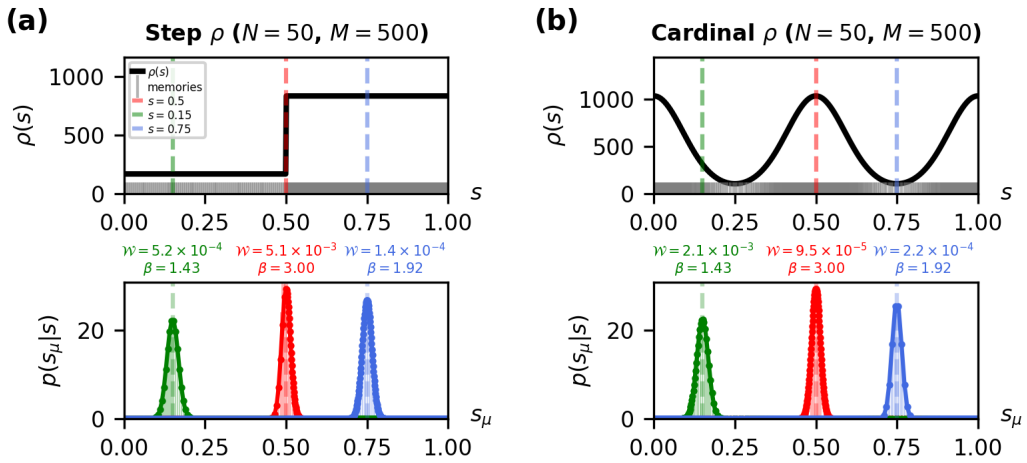


Figure 2: Monte Carlo per-pattern retrieval density ($N = 50$ neurons, $M = 500$ patterns, Gaussian likelihood) versus the local Gaussian approximation Eq. (3), under (a) step and (b) cardinal memory density $\rho(s)$. Wasserstein distance $\mathcal{W}$ below 1%; the approximation degrades slightly where ρ varies rapidly (red in (a) and green in (b)). See Figs. 4 to 10 and Tabs. 3 and 4 for additional variations of this experiment.

Interpretation. The optimal encoder–decoder system allocates retrieval resolution in proportion to memory density, such that $\sqrt{\gamma^*(s)}/\rho^*(s) = C/M$ is spatially uniform. Although the loss exponent p determines how strongly the optimal allocations depend on the stimulus prior, the quadratic coupling between retrieval precision and memory density remains invariant across all L^p losses.

Connection to efficient coding. When memory density is fixed, the L^p objective reduces, up to a constant factor, to the classical efficient-coding distortion criterion, yielding $\gamma^*(s) \propto p(s)^{2/(p+1)}$ under the resolution budget in Eq. (8) [51, 53]. Allowing memory density to be jointly optimized changes the prior-dependent exponent to $2/(p+2)$, while establishing the general coupling between optimal retrieval precision and memory allocation.

4.4 Recall error and capacity

Retrieval error and capacity are the standard performance metrics in the modern Hopfield literature. We analyze them by expressing both in terms of the precision $\gamma(s)$ and memory density $\rho(s)$ introduced in our framework. The full derivation and more detailed explanation are given in Sec. I.

Proposition 2 (Recall error and capacity)**.** Let $\bar{s}$ denote the retrieved value induced by the per-pattern density Eq. (3) weighted by ρ (as in Eq. (4)). Under (A1) and (A2), the mean squared retrieval error is

$$\mathrm{MSE}(s) \;:=\; \mathbb{E}_{p_{\mathrm{ret}}}\big[(\bar{s}-s)^2 \mid s\big] \;\approx\; \frac{1}{\gamma(s)} \;+\; \frac{\rho''(s)}{\gamma(s)^2\rho(s)}. \tag{12}$$

For any tolerance $\varepsilon > 0$, the probability of correct retrieval satisfies

$$\mathbf{P}_{\mathrm{correct}}(s) \;:=\; \Pr\big[|\bar{s}-s| < \varepsilon \mid s\big] \;\geq\; 1 \;-\; \frac{1}{\varepsilon^2}\left[\frac{1}{\gamma(s)} + \frac{\rho''(s)}{\gamma(s)^2\rho(s)}\right]. \tag{13}$$

Proof Sketch. The full proof and discussion can be found in Appendix I. The MSE expansion follows from a second-order Taylor expansion of $\log \rho(\bar{s})$ around s in the density Eq. (4). The capacity bound follows by Markov's inequality applied to $w^2 = (\bar{s}-s)^2$. □

5 Results

We first evaluate the accuracy of the local Gaussian approximation in Eq. (3), which underlies optimal retrieval precision and memory allocation (see Sec. 4). We then compare the optimal $\beta(s)$ in Eq. (10)

against the encoder-blind Bayesian baseline (constant $\beta(s)$) in terms of retrieval accuracy and coding resource. Finally, we derive a closed-form expression for memory recall bias that explains empirically observed biases across sensory domains (e.g., angle, spatial location, and time). All simulations were run on an Apple M2 machine with 16 GB memory and completed within a minute.

5.1 Asymptotic local Gaussian evaluation

To derive the per-pattern retrieval density in Eq. (3), we assume a large number of neurons and stored memory patterns, which may be overly restrictive in realistic regimes. To assess the impact of this assumption, we employ a Monte Carlo approach (40,000 samples) to evaluate the integral in Eq. (1). Specifically, we assume independent Gaussian likelihoods with a finite population size ($N = 50$ neurons) and a finite memory load ($M = 500$ stored patterns), under both step and cardinal memory densities. As a representative non-constant β, we use a truncated Gaussian $\beta(s)$ defined on $[0, 1]$ with a mean of 0.5 and a variance of 0.04. See Sec. F for additional $\beta(s)$ distributions, including bimodal and constant, as well as Poisson likelihoods and additional values of N and M, all of which produce similar results. Fig. 2 shows that the retrieval densities are well approximated by a Gaussian profile in both dense and sparse memory regimes, with Wasserstein distance below 1%. The approximation degrades slightly in regions where the memory density varies rapidly (see the red and green curves for the step (Fig. 2a) and cardinal (Fig. 2b) densities, respectively). This behavior is consistent with the validity condition of the Laplace approximation ($\rho(s)$ must vary on a length scale longer than $1/\sqrt{\beta(s)\mathrm{FI}_{\mathrm{enc}}}$) as assumed in the derivation (see Sec. B).

5.2 Comparison to encoder-blind Bayesian

Hopfield retrieval with a stimulus-independent β is equivalent to the posterior mean under a Gaussian likelihood with a uniform prior (Sec. G). Specifically, it performs Bayesian estimation, where β^{-1} parameterizes the covariance of an isotropic Gaussian noise model. Setting $\beta(s) = 1$ therefore defines a stimulus-independent, encoder-agnostic baseline against which we evaluate the adaptive $\beta^*(s)$. We first consider a Gaussian prior (see Fig. 3a) with constant $\mathrm{FI}_{\mathrm{enc}}$ and $\mathcal{L}_2$ loss (see Sec. H.2 for $\mathcal{L}_1$ loss). More simulation details are in Sec. H.1.

Our framework using the optimal stimulus-dependent $\beta^*(s)$ outperforms the baseline model with $\beta = 1$ but requires more coding resource C (see Fig. 3b). To determine whether this performance gain arises from increased coding resource C, we project the performance of $\beta^*(s)$ onto the family of constant-β models (blue curve in Fig. 3b). For matched performance, the adaptive $\beta^*(s)$ requires less coding resource.

To explain this better-than-Bayesian performance, we illustrate per-stimulus loss under equal coding resource (purple: constant β; green: $\beta^*(s)$). The optimal $\beta^*(s)$ reallocates resources by prioritizing regions with higher stimulus probability, while sacrificing performance in less frequent regions. This redistribution improves overall expected performance.

We further evaluate the derived optimal $\beta^*(s)$ in finite-size synthetic Hopfield retrieval using Eq. (2). The results (see Sec. H.4) demonstrate that the theoretical improvements in retrieval accuracy and MSE persist in finite-memory retrieval and are not solely artifacts of the approximations used in the derivation.

5.3 Memory recall bias

Memory recall biases have been widely documented [3, 4, 24, 25, 56]. Using the per-pattern retrieval density in Eq. (3), we derive the bias predicted by our framework.

Proposition 3 (Memory recall bias)**.** Let $p_{\mathrm{ret}}(\bar{s} \mid s)$ denote the retrieval density induced by the per-pattern density Eq. (3) weighted by the memory density ρ. Under (A1) and (A2), to leading order in the local Taylor expansion of $\log \rho$ around s, the recall bias is

$$b(s) := \mathbb{E}_{p_{\mathrm{ret}}}[\bar{s} - s] = \frac{(\log \rho)'(s)}{\gamma(s)}. \tag{14}$$

Retrieval is attracted toward regions of higher memory density, with magnitude inversely proportional to the retrieval precision. The detail and proof are in Sec. C.

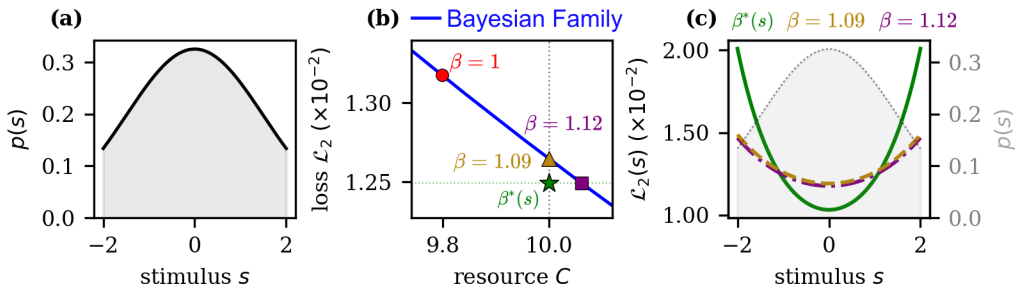


Figure 3: Adaptive $\beta^*(s)$ versus encoder-agnostic Bayesian baseline (constant β) under Gaussian prior, constant $\mathrm{FI}_{\mathrm{enc}} = 12$, and $\mathcal{L}_2$ loss. (a) Prior Gaussian distribution. (b) Global $\mathcal{L}_2$ loss against coding resource C; $\beta^*(s)$ outperforms the baselines in loss (red: $\beta = 1$, yellow: $\beta = 1.09$) or in resource (purple: $\beta = 1.12$). (c) Per-stimulus loss $\mathcal{L}_2(s)$ at matched global $\mathcal{L}_2$ (purple and yellow: constant β; green: $\beta^*(s)$); the adaptive decoder reallocates resources to high-prior regions. Yellow and purple points correspond resource-matched and performance-matched conditions. See Fig. 11 for $\mathcal{L}_1$ loss.

Table 1: Our framework can explain central tendency and cognitive-load effects observed in cognitive psychology across diverse sensory types.

Work	Sensory Type	Effect
Huttenlocher et al., 1991 [24]	Location (angle)	Central tendency
Jazayeri & Shadlen, 2010 [25]	Time	
Ashourian & Loewenstein, 2011 [4]	Length	
Huttenlocher et al., 1991 [24]	Location	Cognitive load ↑ bias ↑
Ashourian & Loewenstein, 2011 [4]	Length	
Allred et al., 2016 [3]	Length	
Zang et al., 2026 [56]	Time	

Connection to Wei-Stocker law [54]. Let $D(s) := \gamma(s)^{-1/2}$ denote the discrimination threshold. Substituting the optimal allocation of Prop. 1 into the bias formula of Prop. 3, and using the general coupling $\gamma^*(s) = \kappa\,[\rho^*(s)]^2$ from Thm. 1 to express both $b^*(s)$ and $(D^2)'(s)$ in terms of ρ^*, a direct calculation yields $b^*(s) = -\frac{1}{2}\frac{d}{ds}D^2(s)$. This is the bias–discriminability law of Wei and Stocker [54]. The constant $1/2$ comes from the general coupling alone, not from the specific form of ρ^* or γ^*. The law therefore holds across all L^p losses with $p > 0$. We would like to emphasize that Wei and Stocker derived this same relation for continuous perceptual estimation under efficient coding. Our derivation uses a different model: discrete memory recall in a Hopfield-type network.

Evidence from psychological experiments. Although $\rho(s)$ and $\gamma(s)$ are not directly observable in psychological experiments, the bias formula Eq. (14) still yields qualitative predictions that can be checked against the literature. First, $(\log \rho)'(s)$ in the numerator predicts that retrieval is attracted toward regions of higher memory density. This matches the central tendency effect reported in [4, 24, 25]. Second, $\gamma(s)$ in the denominator sets the retrieval precision. Cognitive load is known to lower retrieval precision, corresponding to $\gamma(s) \downarrow$ in our model. The bias formula then predicts that the magnitude of the bias grows under higher cognitive load. This is consistent with [3, 4, 24, 56]. A sharper prediction follows. If cognitive load rescales $\gamma(s)$ approximately uniformly across stimuli while leaving $\rho(s)$ unchanged, the spatial shape of the bias should be preserved and only its overall magnitude should shift. This pattern is reported in [3, 24]. We summarize the predictions and the empirical evidence in Tab. 1. Note that the supporting evidence spans diverse stimulus types, indicating that the predictions are not specific to any particular stimulus class.

6 Discussion

We introduce precision-aware Hopfield retrieval, which incorporates the encoder's Fisher information into a Hopfield decoder with stimulus-dependent inverse temperature $\beta(s)$. Joint optimization of memory density $\rho(s)$ and retrieval precision $\gamma(s)$ yields closed-form solutions shaped by the prior and loss (Prop. 1). We also characterize the trade-off between recall error and memory capacity (Prop. 2). L^p losses with different values of p are united under the general relation $\gamma^*(s) \propto [\rho^*(s)]^2$ (Thm. 1). We further derive a closed-form recall bias (Prop. 3) that recovers the Wei–Stocker perception law [54] in the memory recall setting and explains central-tendency and cognitive-load effects on recall across sensory domains.

The primary limitations arise from assumptions (A1) and (A2). Although potentially restrictive from a biological perspective, we evaluate their impact via simulations across varying numbers of neurons N and memory patterns M, Gaussian/Poisson encoding models, and different $\beta(s)$ and $\rho(s)$ in the decoder. We find that deviations from the local Gaussian approximation (measured by Wasserstein distance) remain low. Additionally, we acknowledge that the scalar stimulus setting limits applicability when extended to multidimensional scenarios (e.g., images). However, the approach in [55] provides a natural extension to our framework.

Connection to diffusion models. By approximating the variance of the Hopfield readout $\bar{s}$ given s at leading order as $\mathrm{Var}(\bar{s} \mid s) \approx 1/\gamma(s)$ from Eq. (12), we find an intriguing relation: $(\log \rho)'(s) \approx b(s)/\mathrm{Var}(\bar{s} \mid s)$. This expression can be interpreted as a heuristic connection between denoising and score estimation, where the bias term $b(s)$ corresponds to the expected reconstruction error induced by Gaussian corruption, and $\mathrm{Var}(\bar{s} \mid s)$ plays the role of the noise scale that controls the smoothing of the data distribution.

From this perspective, score-based diffusion models [46] can be viewed as learning a family of such normalized denoising directions across a continuum of noise levels. In particular, at each diffusion time step t, the model estimates the score of the perturbed distribution, where the noise schedule determines the effective variance of the corruption process.

The relation above places precision-aware Hopfield retrieval in the same family as denoising score estimation: Alain & Bengio [2] showed that a denoising auto-encoder recovers the score of a prior via reconstruction residuals, and our formula is the heteroscedastic, memory-allocation analog. From this perspective, it is interesting to explore how the naturally emerging term $(\log \rho)'(s)$ may serve as a general-purpose score signal, consistent with its role in score-based generative modeling. This retrieval-generation connection may provide a mathematical basis for understanding the observed relationship between recall and imagination in cognitive science [16].

Neural mechanisms for learning $\rho(s)$ and $\beta(s)$. Our framework is normative and provides a particular formalization within the broader family of Bayesian encoder–decoder models, linking encoder precision, memory allocation, and decoder temperature to psychological phenomena. Applying the framework first requires estimating the stimulus prior from environmental statistics, and the encoder Fisher information may then be estimated from neuronal spiking data. Such distribution pairs have already been characterized for many features of natural sensory stimuli in both visual and auditory domains [13].

Estimating $\rho(s)$ and $\beta(s)$ is more challenging because they do not directly correspond to observable neural quantities. Nevertheless, our framework provides an indirect empirical route. Retrieval precision $\gamma(s)$ can be estimated from trial-to-trial recall variability or discrimination thresholds, while Eq. (14) implies $(\log \rho)'(s) = \gamma(s)b(s)$. Thus, joint measurements of recall bias $b(s)$ and retrieval precision can constrain $\rho(s)$ up to a normalization constant. Given an independent estimate of $\mathrm{FI}_{\mathrm{enc}}(s)$, the corresponding inverse temperature can then be inferred through Eq. (10). Behavioral and neural measurements across varying experimental conditions could therefore test the predictions and limitations of our framework. Developing a specific neural mechanism for learning these quantities remains an important direction for future work.

See Sec. J for an extended discussion connecting this work to transformers, and computational neuroscience, including novel testable predictions.

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A Notation

Table 2 summarizes the symbols used throughout the paper.

Table 2: Notation used in the paper.

Symbol	Description
Stimulus and population response	
$s \in \mathcal{S}$	Scalar stimulus
$\mathcal{S}$	Stimulus space
$\mathbf{r} \in \mathbb{R}^N$	Population response (firing rates of N neurons)
N	Number of neurons in the encoder population
$\hat{s} := \arg\max_s p(\mathbf{r} \mid s)$	Maximum likelihood estimator of s from $\mathbf{r}$
$\bar{s}$	Retrieved (decoded) stimulus value
$p(s)$	Stimulus prior
Memory	
$\{s_\mu\}_{\mu=1}^M$	Stored memory patterns
M	Number of stored patterns (memory budget)
$\rho(s)$	Continuous memory density (large-M limit of $\{s_\mu\}$)
Encoder	
$p(\mathbf{r} \mid s)$	Encoder likelihood
$\mathrm{FI}_{\mathrm{enc}}(s)$	Fisher information of the encoder at s
$\mathcal{I}(\hat{s})$	Observed Fisher information at $\hat{s}$
Decoder	
$\mathcal{E}(\mathbf{r}, s_\mu) := -\log p(\mathbf{r} \mid s_\mu)$	Decoder energy (negative log-likelihood)
$\beta(\mathbf{r})$	Inverse temperature of the softmax decoder
$\beta(s)$	Stimulus-dependent inverse temperature (large-N limit)
$\gamma(s) = \beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)/(\beta(s)+1)$	Effective retrieval precision
Retrieval densities	
$p(s_\mu \mid \mathbf{r})$	Decoder posterior over patterns given response
$p(s_\mu \mid s)$	Per-pattern retrieval density (Eq. (3))
$p(\hat{s} \mid s)$	Sampling distribution of the MLE
$p_{\mathrm{ret}}(\bar{s} \mid s)$	Readout retrieval density (Eq. (4))
$\bar{s}_{\mathrm{out}}(s) := \mathbb{E}_{p_{\mathrm{ret}}}[\bar{s}]$	Decoded stimulus estimate
Retrieval objectives	
$\mathcal{L}_p(s),\ \mathcal{L}_p$	Pointwise / global L^p loss (Eqs. (6) and (7))
$c_p = 2^{p/2}\Gamma\big(\frac{p+1}{2}\big)/\sqrt{\pi}$	p-th absolute moment of a standard Gaussian
Constraints and optimal allocation	
C	Coding-resource budget, $\int \sqrt{\gamma(s)}\,\mathrm{d}s = C$
$C_{\max} := \int \sqrt{\mathrm{FI}_{\mathrm{enc}}(s)}\,\mathrm{d}s$	Hard ceiling on C from the encoder
C_{sat}	Loss-dependent feasibility threshold (Eq. (47))
$\rho^*(s),\ \gamma^*(s),\ \beta^*(s)$	Optimal memory density, precision, inverse temperature
κ	General coupling constant: $\gamma^*(s) = \kappa[\rho^*(s)]^2$
$\mathfrak{L}_p$	Lagrangians for L^p-loss
$\lambda_\rho,\ \lambda_\gamma$	Lagrange multipliers for M and C budgets
Bias and performance metrics	
$b(s) := \mathbb{E}_{p_{\mathrm{ret}}}[\bar{s} - s]$	Recall bias (Eq. (14))
$D(s) := \gamma(s)^{-1/2}$	Discrimination threshold
$\mathrm{MSE}(s)$	Mean squared retrieval error (Eq. (12))
$\mathbf{P}_{\mathrm{correct}}(s)$	Correct-retrieval probability (Eq. (13))

B Locally Gaussian per-pattern retrieval density

We derive the closed-form expression for the per-pattern retrieval density

$$p(s_\mu \mid s) = \int p(s_\mu \mid \mathbf{r})\, p(\mathbf{r} \mid s)\, d\mathbf{r} \tag{15}$$

stated in Eq. (3), expanding on the sketch in Sec. 3.2. Throughout, we work in the (A1) large population limit $N \to \infty$ and the (A2)dense memory regime.

Step 1: Taylor expansion of the energy. Let $\hat{s} := \arg\max_s p(\mathbf{r} \mid s)$ be the maximum likelihood estimator. A second-order Taylor expansion of $\mathcal{E}(\mathbf{r}, s_\mu) = -\log p(\mathbf{r} \mid s_\mu)$ around $\hat{s}$ gives

$$\mathcal{E}(\mathbf{r}, s_\mu) \approx \mathcal{E}(\mathbf{r}, \hat{s}) + \tfrac{1}{2}\, \mathcal{I}(\hat{s})\, (s_\mu - \hat{s})^2, \tag{16}$$

where $\mathcal{I}(\hat{s}) := -\partial_s^2 \log p(\mathbf{r} \mid s)\big|_{s=\hat{s}}$ is the observed Fisher information; the linear term vanishes by definition of $\hat{s}$. Under (A1), the law of large numbers gives $\mathcal{I}(\hat{s}) \to \mathrm{FI}_{\mathrm{enc}}(\hat{s})$, and asymptotic consistency of the MLE ($\hat{s} \to s$) together with smooth $\mathrm{FI}_{\mathrm{enc}}$ yields $\mathcal{I}(\hat{s}) \approx \mathrm{FI}_{\mathrm{enc}}(s)$. The Hopfield energy is therefore quadratic in s_μ around $\hat{s}$:

$$\mathcal{E}(\mathbf{r}, s_\mu) \approx \mathcal{E}(\mathbf{r}, \hat{s}) + \tfrac{1}{2}\, \mathrm{FI}_{\mathrm{enc}}(s)\, (s_\mu - \hat{s})^2. \tag{17}$$

Step 2: Reduction of $\beta(\mathbf{r})$ via asymptotic sufficiency. The decoder Eq. (2) uses $\beta(\mathbf{r})$. Under (A1), $\hat{s}$ is asymptotically sufficient for $\mathbf{r}$, so we restrict $\beta(\mathbf{r}) = \tilde{\beta}(\hat{s}(\mathbf{r}))$ without loss of generality. Asymptotic consistency, $\hat{s} \to s$, together with smooth $\tilde{\beta}$ then gives $\tilde{\beta}(\hat{s}) \approx \beta(s)$, where $\beta(s) := \tilde{\beta}(s)$. This parallels the treatment of $\mathcal{I}(\hat{s}) \approx \mathrm{FI}_{\mathrm{enc}}(s)$ in Step 1.

Step 3: Partition function in the dense memory regime. Substituting Eq. (17) into the softmax kernel Eq. (2):

$$p(s_\mu \mid \mathbf{r}) \approx \frac{\exp\Big(-\frac{\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)}{2}(s_\mu - \hat{s})^2\Big)}{\sum_{\mu'} \exp\Big(-\frac{\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)}{2}(s_{\mu'} - \hat{s})^2\Big)}. \tag{18}$$

Under (A2), the denominator is approximated by an integral against the memory density ρ:

$$\sum_{\mu'} \exp\Big(-\tfrac{\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)}{2}(s_{\mu'} - \hat{s})^2\Big) \approx \int \rho(u)\, \exp\Big(-\tfrac{\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)}{2}(\hat{s} - u)^2\Big)\, du \tag{19}$$

$$\overset{\text{Laplace}}{\approx} \rho(\hat{s}) \sqrt{\frac{2\pi}{\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)}} \tag{20}$$

$$\overset{\text{MLE}}{\approx} \rho(s) \sqrt{\frac{2\pi}{\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)}}. \tag{21}$$

The Laplace step approximates $\rho(u) \approx \rho(\hat{s})$ within the Gaussian window centered at $\hat{s}$, and the MLE step uses asymptotic consistency $\hat{s} \approx s$. Substituting back into Eq. (18) gives the locally Gaussian decoder:

$$p(s_\mu \mid \mathbf{r}) \approx \frac{1}{\rho(s)} \sqrt{\frac{\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)}{2\pi}}\, \exp\Big(-\tfrac{\beta(s)\,\mathrm{FI}_{\mathrm{enc}}(s)}{2}(s_\mu - \hat{s})^2\Big). \tag{22}$$

Step 4: MLE asymptotic normality. Under (A1), $\hat{s}$ is an asymptotically sufficient statistic for $\mathbf{r}$ (Bernstein–von Mises), so the marginalization reduces to a one-dimensional integral over $\hat{s}$:

$$p(s_\mu \mid s) = \int p(s_\mu \mid \mathbf{r})\, p(\mathbf{r} \mid s)\, d\mathbf{r} \approx \int p(s_\mu \mid \hat{s})\, p(\hat{s} \mid s)\, d\hat{s}. \tag{23}$$

The MLE asymptotic normality gives

$$p(\hat{s} \mid s) \to \mathcal{N}\big(\hat{s};\, s,\, \mathrm{FI}_{\mathrm{enc}}(s)^{-1}\big). \tag{24}$$

Step 5: Convolution. Combining Eqs. (22) and (24):

$$p(s_\mu \mid s) \approx \frac{1}{\rho(s)} \int \mathcal{N}\big(s_\mu;\, \hat{s},\, (\beta(s)\, \mathrm{FI}_{\mathrm{enc}}(s))^{-1}\big)\, \mathcal{N}\big(\hat{s};\, s,\, \mathrm{FI}_{\mathrm{enc}}(s)^{-1}\big)\, \mathrm{d}\hat{s}. \tag{25}$$

The convolution of two independent Gaussians is a Gaussian whose mean and variance are the respective sums:

$$p(s_\mu \mid s) \approx \frac{1}{\rho(s)}\, \mathcal{N}\big(s_\mu;\, s,\, \gamma(s)^{-1}\big)\,, \qquad \gamma(s) = \frac{\beta(s)\, \mathrm{FI}_{\mathrm{enc}}(s)}{\beta(s)+1}, \tag{26}$$

which is Eq. (3).

Remark. Our derivations rely on the large-population and dense-memory assumptions for theoretical tractability. We evaluate the accuracy of the per-pattern retrieval density approximation in Eq. (3) with finite numbers of neurons and stored memories (see Fig. 2). While our simulations suggest that the approximation remains accurate in representative finite-size regimes, finite-N fluctuations and finite-M pattern-spacing effects may introduce deviations.

Specifically, our closed-form theory characterizes the coarse-grained dense-memory regime. When the number of attractors within a local retrieval kernel becomes small, the precise locations and spacings of individual patterns become relevant, and the smooth-density approximation must be replaced by an explicitly discrete analysis.

C Retrieval bias from non-uniform memory density

The retrieval density over stimulus space, obtained by weighting the marginalized retrieval probability Eq. (3) by the memory density, is

$$p_{\mathrm{ret}}(\bar{s} \mid s) = \frac{\rho(\bar{s})}{\rho(s)} \mathcal{N}\big(\bar{s};\, s,\, \gamma(s)^{-1}\big)\,, \tag{27}$$

restating Eq. (4). When ρ varies spatially, the density ratio tilts the retrieval distribution and introduces a systematic bias.

Let $w := \bar{s} - s$. A first-order Taylor expansion of $\log \rho$ around s gives

$$\rho(s + w) \approx \rho(s)\,\big(1 + (\log \rho)'(s)\, w\big), \tag{28}$$

so the tilted density is, to leading order,

$$p_{\mathrm{ret}}(s + w \mid s) \approx \big(1 + (\log \rho)'(s)\, w\big)\, \mathcal{N}\big(w;\, 0,\, \gamma(s)^{-1}\big)\,. \tag{29}$$

The symmetric Gaussian contributes zero to $\mathbb{E}[w]$, and only the linear term survives:

$$\mathbb{E}_{p_{\mathrm{ret}}}[w] \approx (\log \rho)'(s)\, \mathbb{E}\big[w^2\big] = \frac{(\log \rho)'(s)}{\gamma(s)}. \tag{30}$$

The retrieval bias is therefore

$$b(s) := \mathbb{E}_{p_{\mathrm{ret}}}[\bar{s} - s] = \frac{(\log \rho)'(s)}{\gamma(s)}. \tag{31}$$

Retrieval is biased toward regions of higher memory density. Where ρ increases, the retrieved stimulus shifts in that direction. The magnitude is inversely proportional to the precision $\gamma(s)$.

D L^p retrieval loss in closed form

We derive the closed form of the pointwise L^p loss

$$\mathcal{L}_p(s) := \frac{1}{\rho(s)} \int |s_\mu - s|^p \, \rho(s_\mu) \, p(s_\mu \mid s) \, \mathrm{d}s_\mu, \tag{32}$$

stated in Eq. (6). Here, $\rho(s_\mu)$ accounts for the local number of stored patterns in the continuum approximation, while the factor $\rho(s)^{-1}$ explicitly weights retrieval distortion by the inverse local memory density.

Substituting the locally Gaussian per-pattern density Eq. (3), we obtain

$$\mathcal{L}_p(s) \approx \frac{1}{\rho(s)^2} \int |s_\mu - s|^p \, \rho(s_\mu) \, \mathcal{N}\big(s_\mu;\, s,\, \gamma(s)^{-1}\big) \, \mathrm{d}s_\mu \tag{33}$$

$$\approx \frac{1}{\rho(s)} \int |s_\mu - s|^p \, \mathcal{N}\big(s_\mu;\, s,\, \gamma(s)^{-1}\big) \, \mathrm{d}s_\mu, \tag{34}$$

where the second approximation uses the locally uniform memory density assumption $\rho(s_\mu) \approx \rho(s)$ within the Gaussian retrieval window.

Changing variables to $u := s_\mu - s$ gives

$$\mathcal{L}_p(s) \approx \frac{1}{\rho(s)} \int |u|^p \, \mathcal{N}\big(u;\, 0,\, \gamma(s)^{-1}\big) \, \mathrm{d}u \tag{35}$$

$$= \frac{1}{\rho(s)} \, \mathbb{E}\big[|U|^p\big], \tag{36}$$

where $U \sim \mathcal{N}\big(0, \gamma(s)^{-1}\big)$.

Write $U = \sigma Z$ with $Z \sim \mathcal{N}(0,1)$ and $\sigma = \gamma(s)^{-1/2}$. The p-th absolute moment scales as $\mathbb{E}[|U|^p] = \sigma^p \mathbb{E}[|Z|^p]$. The standard-normal moment is

$$\mathbb{E}[|Z|^p] = \frac{2}{\sqrt{2\pi}} \int_0^\infty z^p e^{-z^2/2} \, \mathrm{d}z \tag{37}$$

$$\overset{t=z^2/2}{=} \frac{2^{p/2}}{\sqrt{\pi}} \Gamma\left(\frac{p+1}{2}\right) =: c_p, \tag{38}$$

where the substitution $t = z^2/2$ reduces the integral to the gamma function.

Combining these results yields

$$\mathcal{L}_p(s) \approx \frac{c_p}{\rho(s)} \, \gamma(s)^{-p/2}, \tag{39}$$

which is the locally Gaussian approximation in Eq. (6).

Remark. To obtain an analytically tractable objective, we approximate the discrete sum over stored patterns by a continuous integral. Strictly speaking, this approximation requires weighting the per-pattern retrieval probability by the local memory density $\rho(s_\mu)$, which cancels the factor $\rho(s)^{-1}$ under the locally uniform density assumption. To explicitly account for memory allocation, we introduce an additional factor $\rho(s)^{-1}$, thereby defining a density-weighted retrieval distortion that penalizes sparse memory regions.

Remark. Interestingly, classical efficient coding establishes a connection between logarithmic and L^p losses in the limit $p \to 0^+$ under suitable regularity conditions [7, 50, 53]. However, this connection does not directly extend to our framework, as the additional memory-density weighting $\rho(s)^{-1}$ is independent of p. Consequently, the resulting optimization problem is not guaranteed to be convex in $\rho(s)$.

E Lagrangian derivation of the optimal allocation

We derive the optimal ρ^* and γ^* in Eq. (9) by minimizing Eq. (7) subject to the budgets Eq. (8).

E.1 L^p loss

Form the Lagrangian

$$\mathfrak{L}_p = c_p \int p(s)\, \frac{\gamma(s)^{-p/2}}{\rho(s)}\, \mathrm{d}s + \lambda_\rho \left(\int \rho(s)\, \mathrm{d}s - M \right) + \lambda_\gamma \left(\int \sqrt{\gamma(s)}\, \mathrm{d}s - C \right). \tag{40}$$

Stationarity in ρ. $\delta\mathfrak{L}_p/\delta\rho(s) = -c_p\, p(s)\, \gamma(s)^{-p/2}/\rho(s)^2 + \lambda_\rho = 0$ gives

$$\rho(s)^2 \propto p(s)\, \gamma(s)^{-p/2}. \tag{41}$$

Stationarity in γ. $\delta\mathfrak{L}_p/\delta\gamma(s) = -\frac{p}{2}\, c_p\, p(s)\, \gamma(s)^{-p/2-1}/\rho(s) + \lambda_\gamma/(2\sqrt{\gamma(s)}) = 0$ gives

$$\gamma(s)^{(p+1)/2} \propto \frac{p(s)}{\rho(s)}. \tag{42}$$

Decoupling. From Eq. (41), $\rho \propto p(s)^{1/2}\, \gamma^{-p/4}$. Substituting into Eq. (42),

$$\gamma^{(p+1)/2} \propto \frac{p(s)}{p(s)^{1/2}\, \gamma^{-p/4}} = p(s)^{1/2}\, \gamma^{p/4}. \tag{43}$$

Collecting powers of γ on the left, $\gamma^{(p+1)/2-p/4} = \gamma^{(p+2)/4} \propto p(s)^{1/2}$, which yields

$$\gamma^*_{L^p}(s) \propto p(s)^{\frac{2}{p+2}}. \tag{44}$$

Backsubstituting into $\rho \propto p(s)^{1/2}\, \gamma^{-p/4}$,

$$\rho^*_{L^p}(s) \propto p(s)^{1/2}\, p(s)^{-\frac{p}{2(p+2)}} = p(s)^{\frac{1}{p+2}}. \tag{45}$$

The proportionality constants are fixed by the budgets Eq. (8).

E.2 Feasibility threshold C_{sat}

The interior solutions above do not enforce the pointwise ceiling $\gamma(s) \le \mathrm{FI}_{\mathrm{enc}}(s)$ inherited from Eq. (3). We compute the largest budget C for which the interior solution stays strictly below the ceiling at every s. Restricting C to this range, as we do in the main text, prevents encoder saturation entirely.

L^p loss. Let $Z_p := \int p(s)^{1/(p+2)}\, \mathrm{d}s$. Normalizing $\sqrt{\gamma^*_{L^p}(s)} \propto p(s)^{1/(p+2)}$ against the budget gives

$$\gamma^*_{L^p}(s) = \left(\frac{C}{Z_p} \right)^2 p(s)^{2/(p+2)}, \tag{46}$$

and pointwise feasibility is equivalent to

$$C < C^{L^p}_{\mathrm{sat}} := Z_p \min_s \frac{\sqrt{\mathrm{FI}_{\mathrm{enc}}(s)}}{p(s)^{1/(p+2)}}. \tag{47}$$

Boundary case. For C at or above C_{sat}, the interior closed form would violate the ceiling on a subset of stimulus space; on that subset $\gamma^*(s) = \mathrm{FI}_{\mathrm{enc}}(s)$ and Eq. (10) sends $\beta^*(s) \to \infty$, the zero-temperature decoder. We do not consider this regime in the main text.

F Evaluation of local Gaussian approximation

We evaluate the local Gaussian approximation in Eq. (3) across different numbers of neurons N and memory patterns M (Figs. 4 to 6). We repeat the same experiment under a Poisson likelihood (Figs. 7 to 10). While the approximation degrades slightly for smaller N and M, the Wasserstein distances remain small. We also repeat the experiment for additional β settings and summarize the results in Tabs. 3 and 4. Further implementation details can be found in the accompanying open-source codebase.

Table 3: Wasserstein-1 distances as in Fig. 2 across different settings by using **Step** memory density. The settings are across likelihood, N, M, and β shapes.

Setup	$s = 0.15$	$s = 0.5$	$s = 0.75$
gauss_N20_M100_beta_const	8.55×10^{-4}	8.16×10^{-3}	2.15×10^{-4}
gauss_N20_M500_beta_const	6.59×10^{-4}	8.18×10^{-3}	2.28×10^{-4}
gauss_N50_M100_beta_const	5.91×10^{-4}	4.69×10^{-3}	6.95×10^{-5}
gauss_N50_M500_beta_const	4.36×10^{-4}	5.09×10^{-3}	1.08×10^{-4}
poisson_N20_M100_beta_const	2.97×10^{-3}	1.08×10^{-2}	1.22×10^{-3}
poisson_N20_M500_beta_const	2.35×10^{-3}	1.08×10^{-2}	1.38×10^{-3}
poisson_N50_M100_beta_const	1.10×10^{-3}	6.64×10^{-3}	6.42×10^{-4}
poisson_N50_M500_beta_const	9.61×10^{-4}	6.82×10^{-3}	7.18×10^{-4}
gauss_N20_M100_beta_gauss	1.41×10^{-3}	8.07×10^{-3}	3.39×10^{-4}
gauss_N20_M500_beta_gauss	1.36×10^{-3}	8.27×10^{-3}	3.55×10^{-4}
gauss_N50_M100_beta_gauss	4.96×10^{-4}	4.75×10^{-3}	1.16×10^{-4}
gauss_N50_M500_beta_gauss	5.62×10^{-4}	5.07×10^{-3}	1.49×10^{-4}
poisson_N20_M100_beta_gauss	2.81×10^{-3}	1.10×10^{-2}	1.56×10^{-3}
poisson_N20_M500_beta_gauss	2.71×10^{-3}	1.06×10^{-2}	1.52×10^{-3}
poisson_N50_M100_beta_gauss	1.02×10^{-3}	6.85×10^{-3}	6.22×10^{-4}
poisson_N50_M500_beta_gauss	1.03×10^{-3}	6.78×10^{-3}	4.50×10^{-4}
gauss_N20_M100_beta_bimodal	1.39×10^{-3}	7.18×10^{-3}	2.93×10^{-4}
gauss_N20_M500_beta_bimodal	1.19×10^{-3}	7.21×10^{-3}	3.05×10^{-4}
gauss_N50_M100_beta_bimodal	5.06×10^{-4}	4.22×10^{-3}	1.24×10^{-4}
gauss_N50_M500_beta_bimodal	5.06×10^{-4}	4.49×10^{-3}	1.17×10^{-4}
poisson_N20_M100_beta_bimodal	2.53×10^{-3}	9.55×10^{-3}	1.56×10^{-3}
poisson_N20_M500_beta_bimodal	2.60×10^{-3}	9.34×10^{-3}	1.50×10^{-3}
poisson_N50_M100_beta_bimodal	1.14×10^{-3}	5.83×10^{-3}	5.85×10^{-4}
poisson_N50_M500_beta_bimodal	1.04×10^{-3}	5.93×10^{-3}	6.20×10^{-4}

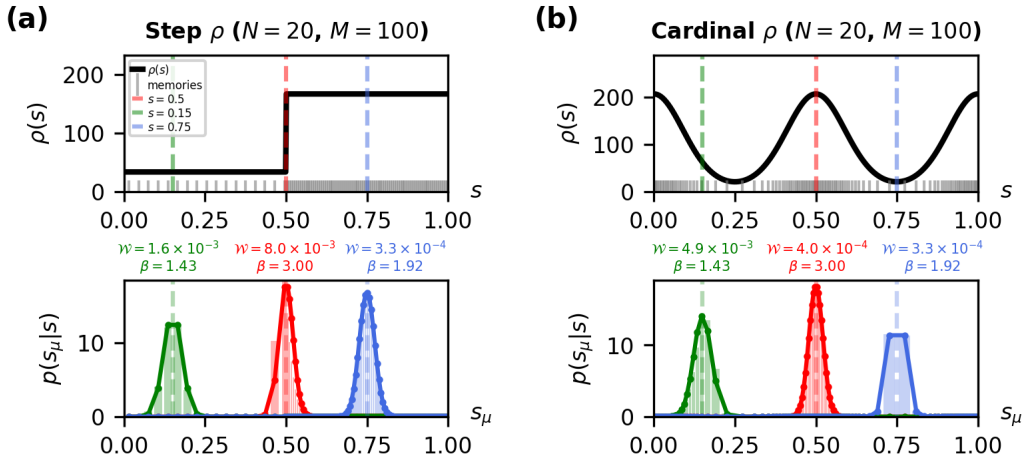


Figure 4: Same as in Fig. 2, but with $N = 20$ and $M = 100$.

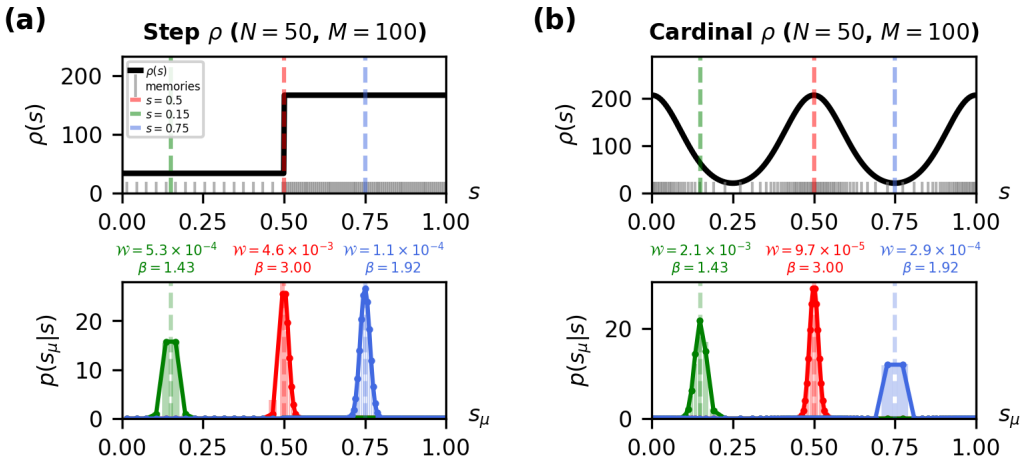


Figure 5: Same as in Fig. 2, but with $N = 50$ and $M = 100$.

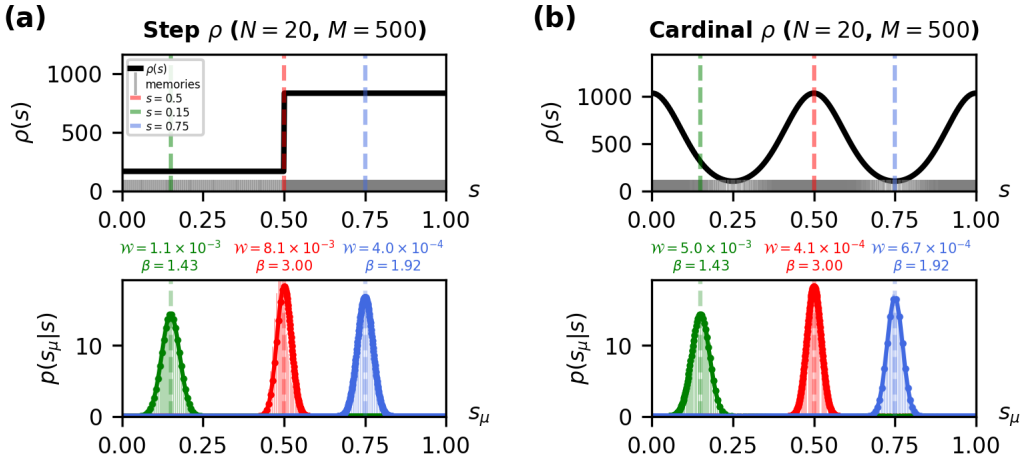


Figure 6: Same as in Fig. 2, but with $N = 20$ and $M = 500$.

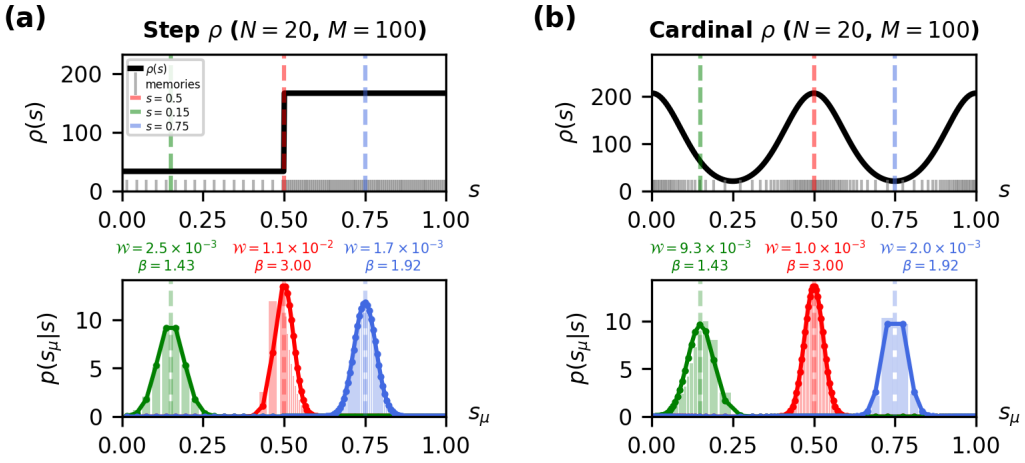


Figure 7: Same as in Fig. 2, but under Poisson likelihood with $N = 20$ and $M = 100$.

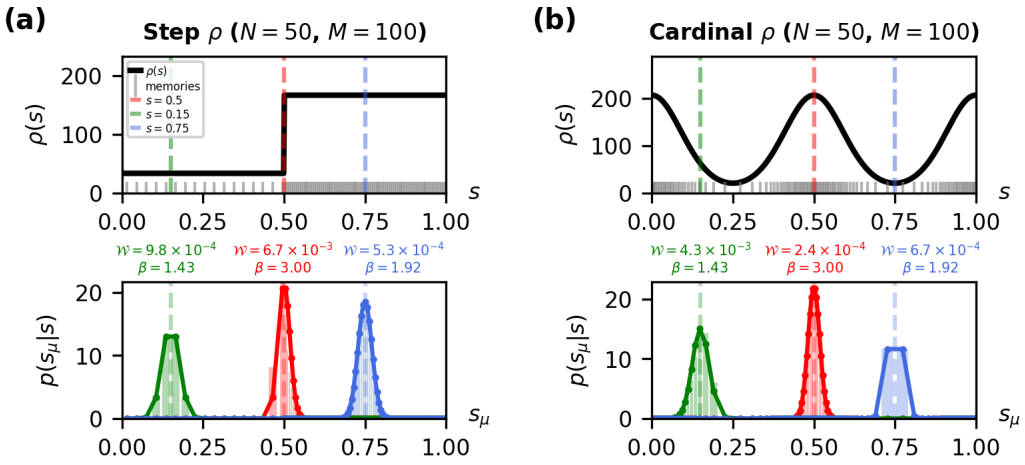


Figure 8: Same as in Fig. 2, but under Poisson likelihood with $N = 50$ and $M = 100$.

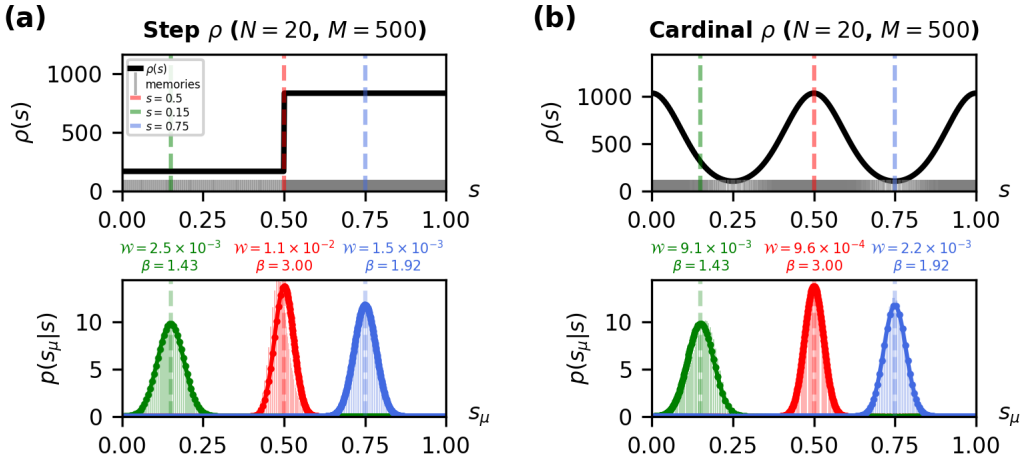


Figure 9: Same as in Fig. 2, but under Poisson likelihood with $N = 20$ and $M = 500$.

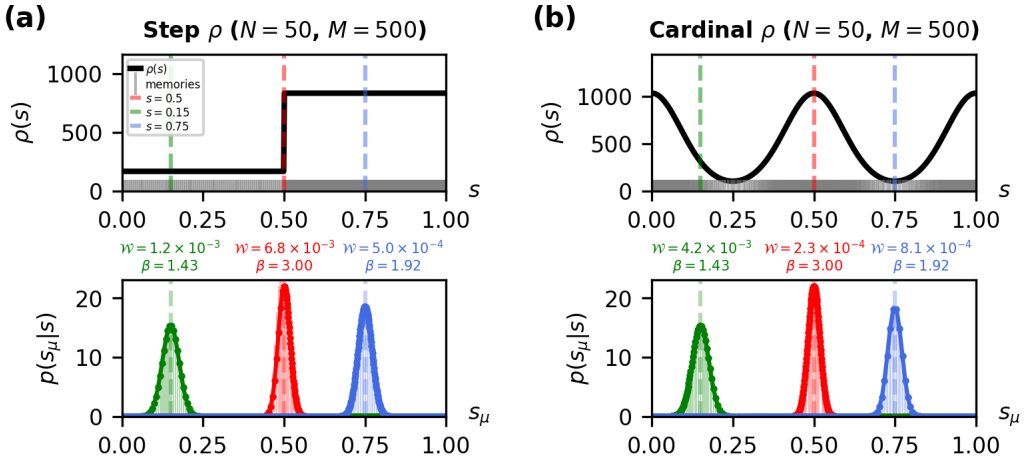


Figure 10: Same as in Fig. 2, but under Poisson likelihood with $N = 50$ and $M = 500$.

Table 4: Wasserstein-1 distances as in Fig. 2 across different settings by using **Cardinal** memory density. The settings are across likelihood, N, M, and β shapes.

Setup	$s = 0.15$	$s = 0.5$	$s = 0.75$
gauss_N20_M100_beta_const	5.43×10^{-3}	3.55×10^{-4}	3.47×10^{-4}
gauss_N20_M500_beta_const	5.27×10^{-3}	3.81×10^{-4}	6.45×10^{-4}
gauss_N50_M100_beta_const	2.36×10^{-3}	1.07×10^{-4}	3.02×10^{-4}
gauss_N50_M500_beta_const	2.29×10^{-3}	7.55×10^{-5}	2.18×10^{-4}
poisson_N20_M100_beta_const	9.29×10^{-3}	8.30×10^{-4}	1.82×10^{-3}
poisson_N20_M500_beta_const	9.85×10^{-3}	7.91×10^{-4}	2.05×10^{-3}
poisson_N50_M100_beta_const	4.28×10^{-3}	2.02×10^{-4}	6.24×10^{-4}
poisson_N50_M500_beta_const	4.35×10^{-3}	1.61×10^{-4}	6.67×10^{-4}
gauss_N20_M100_beta_gauss	4.92×10^{-3}	3.46×10^{-4}	4.53×10^{-4}
gauss_N20_M500_beta_gauss	4.78×10^{-3}	3.23×10^{-4}	6.05×10^{-4}
gauss_N50_M100_beta_gauss	1.98×10^{-3}	1.19×10^{-4}	1.38×10^{-4}
gauss_N50_M500_beta_gauss	2.07×10^{-3}	9.74×10^{-5}	1.85×10^{-4}
poisson_N20_M100_beta_gauss	8.94×10^{-3}	7.05×10^{-4}	2.27×10^{-3}
poisson_N20_M500_beta_gauss	9.26×10^{-3}	8.67×10^{-4}	2.23×10^{-3}
poisson_N50_M100_beta_gauss	4.29×10^{-3}	2.79×10^{-4}	6.91×10^{-4}
poisson_N50_M500_beta_gauss	4.29×10^{-3}	2.73×10^{-4}	7.35×10^{-4}
gauss_N20_M100_beta_bimodal	4.96×10^{-3}	4.58×10^{-4}	2.26×10^{-4}
gauss_N20_M500_beta_bimodal	5.13×10^{-3}	3.18×10^{-4}	6.74×10^{-4}
gauss_N50_M100_beta_bimodal	2.14×10^{-3}	6.91×10^{-5}	4.51×10^{-4}
gauss_N50_M500_beta_bimodal	2.17×10^{-3}	6.73×10^{-5}	1.76×10^{-4}
poisson_N20_M100_beta_bimodal	9.58×10^{-3}	6.83×10^{-4}	1.88×10^{-3}
poisson_N20_M500_beta_bimodal	9.52×10^{-3}	6.88×10^{-4}	2.08×10^{-3}
poisson_N50_M100_beta_bimodal	4.38×10^{-3}	2.05×10^{-4}	7.07×10^{-4}
poisson_N50_M500_beta_bimodal	4.22×10^{-3}	1.83×10^{-4}	7.27×10^{-4}

G Hopfield retrieval as a Bayesian posterior mean

We recap the Bayesian interpretation of modern Hopfield retrieval. This is the standard motivation for the canonical $\beta = 1$ setting, which we use as the encoder-blind baseline in the main text.

Setup. Consider M stored patterns $\{s_\mu\}_{\mu=1}^M \subset \mathbb{R}^d$ and a query $s \in \mathbb{R}^d$. The continuous Hopfield update with L^2 similarity is

$$T_\beta(s) \;=\; \sum_{\mu=1}^{M} s_\mu \cdot \frac{\exp\Big(-\frac{\beta}{2}\|s - s_\mu\|^2\Big)}{\sum_{\nu=1}^{M} \exp\Big(-\frac{\beta}{2}\|s - s_\nu\|^2\Big)}, \tag{48}$$

where $\beta > 0$ controls the sharpness of the softmax. This is the canonical instance of the softmax decoder (analogy to Eq. (2) in the main text), with energy $\mathcal{E}(s, s_\mu) = \frac{1}{2}\|s - s_\mu\|^2$ and a constant inverse temperature.

Generative model. Let ξ denote the random stored pattern with a uniform prior, $\xi \sim \mathrm{Unif}(\{s_1, \ldots, s_M\})$. Treat the query as a noisy Gaussian observation of the chosen pattern,

$$s \mid \xi \;\sim\; \mathcal{N}\big(\xi,\, \beta^{-1} I\big)\,. \tag{49}$$

The covariance is $\beta^{-1} I$, so β is the precision of the observation channel. Bayes' rule cancels the uniform prior and the Gaussian normalization, leaving the posterior mass

$$p(\xi = s_\mu \mid s) \;=\; \frac{\exp\Big(-\frac{\beta}{2}\|s - s_\mu\|^2\Big)}{\sum_{\nu=1}^{M} \exp\Big(-\frac{\beta}{2}\|s - s_\nu\|^2\Big)}. \tag{50}$$

This is exactly the softmax kernel in Eq. (48).

Proposition 4 (Hopfield retrieval is the MMSE estimator)**.** Under the generative model Eq. (49), the Hopfield update Eq. (48) coincides with the posterior mean of the stored patterns:

$$T_\beta(s) \;=\; \mathbb{E}[\xi \mid s] \;=\; \arg\min_{\hat{\xi}(\cdot)} \mathbb{E}\Big[\|\hat{\xi}(s) - \xi\|^2\Big]\,. \tag{51}$$

Proof. The posterior mean of a discrete random variable is $\mathbb{E}[\xi \mid s] = \sum_\mu s_\mu\, p(\xi = s_\mu \mid s)$. Substituting Eq. (50) recovers Eq. (48). The MMSE characterization is the standard variational property of the conditional expectation under squared loss. □

Two roles of β. The same β plays two roles. On the decoder side, it is the softmax inverse temperature. On the generative side, it is the precision of the assumed observation channel. Large β encodes low-noise observations and concentrates the posterior on the nearest pattern. Small β encodes high-noise observations and flattens the posterior toward the uniform prior. The case $\beta = 1$ corresponds to unit observation noise ($\sigma^2 = 1$). It is the canonical reference: stimulus-independent, and using no information about the encoder.

Encoder-blind baseline. This motivates our baseline. The proposed adaptive precision $\beta^*(s)$ is constructed from the local Fisher information $\mathrm{FI}_{\mathrm{enc}}(s)$ of the encoder. It therefore varies across stimulus space. The constant choice $\beta(s) \equiv 1$ ignores $\mathrm{FI}_{\mathrm{enc}}(s)$. It treats every stimulus as if observed through a homogeneous unit-noise channel. We adopt $\beta(s) \equiv 1$ as the *encoder-blind* baseline. $\beta^*(s)$ is evaluated against this baseline.

H Additional results

Here, we provide additional results comparing our framework with Bayesian decoding under L^1 loss (Sec. H.2). In Sec. H.4, we further evaluate our framework using finite-size synthetic Hopfield retrieval based on Eq. (2), comparing it against a resource-matched constant-β baseline and the encoder-blind baseline ($\beta = 1$).

H.1 Simulation details

We compared optimal and constant-β decoding strategies on a one-dimensional stimulus domain $s \in [-2, 2]$, with memory budget $M = 50$ and coding resource $C = 10$. For each condition, we computed the theoretical optimal allocation (ρ^*, γ^*) and evaluated constant-β alternatives while holding memory density fixed at ρ^*. Numerical integrals were evaluated using the trapezoidal rule on 4,001 evenly spaced stimulus points. Constant β (i.e., blue line: Bayesian family in Figs. 3 and 11 to 13) was swept over 400 logarithmically spaced values from 10^{-3} to 10^5. Comparisons included the $\beta = 1$ baseline, a resource-matched constant decoder, and a loss-matched constant decoder. Budget constraints and feasibility, $\gamma^*(s) < \mathrm{FI}(s)$, were checked numerically. The stimulus prior, loss function, and Fisher-information profile were specified separately for each case in Secs. H.2 and H.3.

H.2 Comparison to Bayesian for L^1 losses

We analyze the performance of our framework under L^1 loss. Our framework outperforms others (see Fig. 11). Additionally, it demonstrates that the adaptive decoder reallocates resources to high-prior regions.

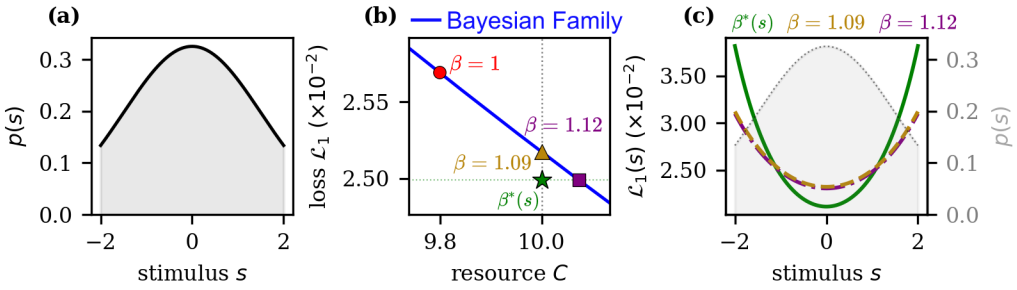


Figure 11: Same as Fig. 3, but under L^1 loss.

H.3 Comparison under bimodal prior

We extend our analysis to a more general setting with a bimodal prior and either uniform or Gaussian $\mathrm{FI}_{\mathrm{enc}}(s)$ (see Figs. 12 and 13). These results further demonstrate that our framework outperforms the baselines.

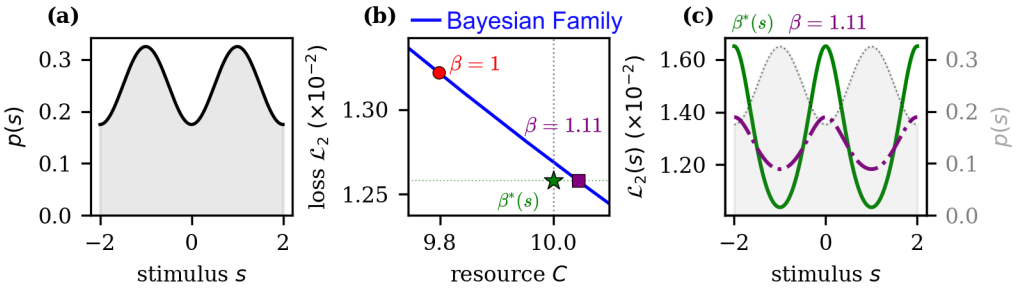


Figure 12: Same as Fig. 3, but with the bimodal prior.

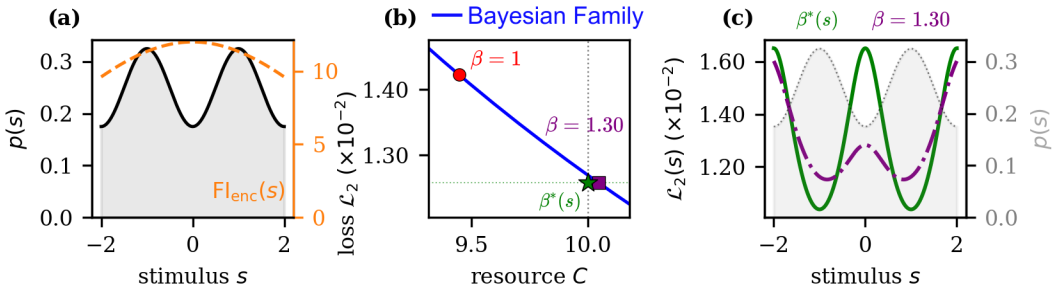


Figure 13: Same as Fig. 3, but with the bimodal prior and Gaussian $\mathrm{FI}_{\mathrm{enc}}(s)$.

Table 5: Performance comparison of finite-size synthetic Hopfield retrieval under different decoding strategies. The adaptive decoder achieves the highest retrieval accuracy and lowest MSE under the same resource budget.

Decoder	Resource	Accuracy ↑	MSE ↓	Mean $\lvert b(s)\rvert$ ↓
Adaptive $\beta^*(\hat{s})$	10.000	**0.5470**	**0.13801**	0.04080
Resource-matched constant $\beta = 1.087$	10.000	0.5412	0.14404	**0.03248**
Encoder-blind $\beta = 1$	9.798	0.5328	0.14967	0.03453

H.4 Finite synthetic Hopfield retrieval

We include a finite one-dimensional retrieval experiment (based on Eq. (2)) with $M = 50$ memories on $s \in [-2, 2]$, a truncated Gaussian prior, and similar other conditions as in Fig. 3. Memories were placed at quantiles of the L^2-optimal density $\rho^*(s) \propto p(s)^{1/4}$. For each target memory, queries were sampled as $q \mid s \sim \mathcal{N}(s, \mathrm{FI}_{\mathrm{enc}}^{-1})$, and finite softmax retrieval used the theoretically derived $\beta^*(\hat{s})$, evaluated at the noisy MLE rather than the true stimulus.

We compared it with $\beta = 1$ and a constant β tuned on 10,000 validation queries subject to the same resource budget. Results were evaluated on 100,000 independent queries. Accuracy is the probability of recall within $\epsilon = \mathrm{FI}_{\mathrm{enc}}^{-1/2}$; error is the expected squared retrieval error; bias is the prior-weighted absolute conditional bias.

At matched resource, the adaptive decoder increased accuracy by 0.58% and reduced MSE by 4.19% (see Tab. 5). Note that the modest improvement in accuracy is expected because the evaluated setting uses $\rho(s)$ and $\beta(s)$ optimized under the L^2 objective, which is optimal only for minimizing MSE. These improvements occurred for all ten tested random seeds. The bias remained directed toward the prior mode, consistent with the predicted central-tendency effect. Its magnitude is a consequence of precision reallocation rather than an optimized objective. These results demonstrate that the theoretical improvement survives finite-memory retrieval and is not solely an artifact of the continuum approximation.

I Proof of Proposition 2

We first restate the setup and Proposition 2. Next, we prove the proposition.

The goal of the memory system is to take a stimulus s and retrieve a stored pattern $\bar{s}$ that is close to s. The retrieval error is $w := \bar{s} - s$. The mean squared error decomposes as

$$\mathrm{MSE}(s) = \mathbb{E}_{p_{\mathrm{ret}}}\big[(\bar{s} - s)^2 \mid s\big] = \mathrm{Var}_{p_{\mathrm{ret}}}(\bar{s} \mid s) + \big[\underbrace{\mathbb{E}_{p_{\mathrm{ret}}}[\bar{s} - s \mid s]}_{\text{bias } b(s)}\big]^2. \tag{52}$$

Proposition 5 (Recall error and capacity)**.** Let $\bar{s}$ denote the retrieved value induced by the per-pattern density Eq. (3) weighted by ρ (as in Eq. (4)). Under (A1) and (A2), the mean squared retrieval error is

$$\mathrm{MSE}(s) \;:=\; \mathbb{E}_{p_{\mathrm{ret}}}\big[(\bar{s} - s)^2 \mid s\big] \;\approx\; \frac{1}{\gamma(s)} \;+\; \frac{\rho''(s)}{\gamma(s)^2 \rho(s)}.$$

For any tolerance $\varepsilon > 0$, the probability of correct retrieval satisfies

$$\mathbf{P}_{\mathrm{correct}}(s) \;:=\; \Pr\big[|\bar{s} - s| < \varepsilon \mid s\big] \;\geq\; 1 \;-\; \frac{1}{\varepsilon^2}\left[\frac{1}{\gamma(s)} + \frac{\rho''(s)}{\gamma(s)^2 \rho(s)}\right].$$

Proof of Proposition 2. From Equation 4, the readout retrieval density is the per-pattern retrieval density weighted by memory density:

$$p_{\mathrm{ret}}(\bar{s} \mid s) = \frac{\rho(\bar{s})}{\rho(s)}\, \mathcal{N}\big(\bar{s};\, s,\, \gamma(s)^{-1}\big),$$

where $\gamma(s) = \dfrac{\beta(s)\,\mathrm{FI}(s)}{\beta(s) + 1}$ is the retrieval precision.

Substituting $w = \bar{s} - s$, the density over w is

$$p(w \mid s) = \frac{\rho(s + w)}{\rho(s)}\, \mathcal{N}\big(w;\, 0,\, \gamma(s)^{-1}\big). \tag{53}$$

When ρ is constant, the ratio is unity, the distribution is a zero-mean Gaussian, and $\mathrm{MSE}(s) = 1/\gamma(s)$. The interesting case is non-uniform ρ, which tilts the Gaussian and introduces bias.

Define the shorthand

$$a := (\log \rho)'(s), \qquad b := (\log \rho)''(s).$$

Taylor-expand $\log \rho$ around s:

$$\log \rho(s + w) \approx \log \rho(s) + a\, w + \tfrac{1}{2}\, b\, w^2.$$

Exponentiating both sides,

$$\frac{\rho(s + w)}{\rho(s)} = \exp\Big(a\, w + \tfrac{1}{2}\, b\, w^2\Big).$$

Expanding the exponential to second order in w:

$$\begin{aligned}\exp\Big(a\, w + \tfrac{1}{2}\, b\, w^2\Big) &\approx 1 + \Big(a\, w + \tfrac{1}{2}\, b\, w^2\Big) + \frac{1}{2}\Big(a\, w + \tfrac{1}{2}\, b\, w^2\Big)^2 \\ &\approx 1 + a\, w + \tfrac{1}{2}\,(a^2 + b)\, w^2,\end{aligned} \tag{54}$$

where in the last line we kept only terms up to $O(w^2)$.

Substituting equation 54 into equation 53 gives

$$p(w \mid s) \approx \Big[1 + a\, w + \tfrac{1}{2}(a^2 + b)\, w^2\Big]\, \mathcal{N}\big(w;\, 0,\, \gamma^{-1}\big), \tag{55}$$

where we write $\gamma \equiv \gamma(s)$ for brevity.

We compute the moments using the standard Gaussian moment identities for $W \sim \mathcal{N}(0,\, \gamma^{-1})$:

$$\mathbb{E}[W] = 0, \qquad \mathbb{E}[W^2] = \gamma^{-1}, \qquad \mathbb{E}[W^3] = 0, \qquad \mathbb{E}[W^4] = 3\,\gamma^{-2}.$$

$$\int p(w \mid s)\, dw \approx 1 + a \cdot 0 + \tfrac{1}{2}(a^2 + b) \cdot \gamma^{-1}$$
$$= 1 + \frac{a^2 + b}{2\gamma}.$$

This departs from unity at $O(\gamma^{-1})$, which is consistent with the approximation order. Define the normalization constant $Z := 1 + \frac{a^2+b}{2\gamma}$ for the computations below.

$$\mathbb{E}[w] = \frac{1}{Z} \int w\, p(w \mid s)\, dw$$
$$= \frac{1}{Z}\Big[\underbrace{\mathbb{E}[W]}_{0} + a\, \underbrace{\mathbb{E}[W^2]}_{\gamma^{-1}} + \tfrac{1}{2}(a^2 + b)\, \underbrace{\mathbb{E}[W^3]}_{0}\Big]$$
$$= \frac{a/\gamma}{1 + (a^2 + b)/(2\gamma)}.$$

To leading order in $1/\gamma$,

$$b(s) := \mathbb{E}_{p_{\text{ret}}}[\bar{s} - s] \approx \frac{(\log \rho)'(s)}{\gamma(s)}. \tag{56}$$

The bias points toward regions where memory density ρ is increasing.

$$\mathbb{E}[w^2] = \frac{1}{Z} \int w^2\, p(w \mid s)\, dw$$
$$= \frac{1}{Z}\Big[\underbrace{\mathbb{E}[W^2]}_{\gamma^{-1}} + a\, \underbrace{\mathbb{E}[W^3]}_{0} + \tfrac{1}{2}(a^2 + b)\, \underbrace{\mathbb{E}[W^4]}_{3\gamma^{-2}}\Big]$$
$$= \frac{\gamma^{-1} + \frac{3(a^2+b)}{2\gamma^2}}{1 + \frac{a^2+b}{2\gamma}}. \tag{57}$$

Expanding to first subleading order,

$$\mathbb{E}[w^2] \approx \frac{1}{\gamma} + \frac{a^2 + b}{\gamma^2}.$$

$$\mathrm{Var}(w) = \mathbb{E}[w^2] - \big(\mathbb{E}[w]\big)^2$$
$$\approx \left[\frac{1}{\gamma} + \frac{a^2 + b}{\gamma^2}\right] - \frac{a^2}{\gamma^2}$$
$$= \frac{1}{\gamma} + \frac{b}{\gamma^2}.$$

That is,

$$\mathrm{Var}_{p_{\text{ret}}}(\bar{s} \mid s) \approx \frac{1}{\gamma(s)} + \frac{(\log \rho)''(s)}{\gamma(s)^2}. \tag{58}$$

The leading term $1/\gamma$ is the irreducible Gaussian noise. The correction $(\log \rho)''/\gamma^2$ captures how the curvature of $\log \rho$ either concentrates or disperses the retrieval distribution.

Combining equation 56 and equation 58:

$$\mathrm{MSE}(s) \approx \frac{1}{\gamma(s)} + \frac{(\log \rho)''(s)}{\gamma(s)^2} + \frac{\big[(\log \rho)'(s)\big]^2}{\gamma(s)^2}.$$

This can be written compactly as

$$\mathrm{MSE}(s) \approx \frac{1}{\gamma(s)} + \frac{(\log \rho)''(s) + \big[(\log \rho)'(s)\big]^2}{\gamma(s)^2} = \frac{1}{\gamma(s)} + \frac{\rho''(s)/\rho(s)}{\gamma(s)^2}, \tag{59}$$

where the last equality uses the identity $(\log \rho)'' + [(\log \rho)']^2 = \rho''/\rho$. Finally, by substituting equation 59 with the definition of $\gamma(s)$, we get

$$\mathrm{MSE}(s) \approx \frac{\beta(s)+1}{\beta(s)\mathrm{FI}(s)} + \frac{\rho(s)''}{\rho(s)} \left(\frac{1}{\beta(s)\mathrm{FI}(s)} \right)^2.$$

Intuitively, the choice of $\beta(s)$ affects the reconstruction error. Second, when the memory density $\rho(s)$ around the stimulus is sparse, the reconstruction error also increases.

Now consider the definition of a success recall with some tolerance $\varepsilon > 0$[1]: Specifically, we obtain the following probability of correct retrieval from L_1 error to squared error

$$\mathbf{P}\left[|w| \geq \varepsilon \mid s\right] = \mathbf{P}\left[w^2 \geq \varepsilon^2 \mid s\right] \leq \frac{\mathbb{E}[w^2 \mid s]}{\varepsilon^2} = \frac{\mathrm{MSE}(s)}{\varepsilon^2}, \tag{60}$$

where the last inequality comes from Markov's inequality.

Therefore, the correct recall rate is defined as

$$\mathbf{P}_{\mathrm{correct}}(s) = \mathbf{P}\left[|w| < \varepsilon \mid s\right] \geq 1 - \frac{\mathrm{MSE}(s)}{\varepsilon^2}. \tag{61}$$

By direct substitution, we get

$$\mathbf{P}_{\mathrm{correct}}(s) \geq 1 - \frac{1}{\varepsilon^2}\left[\frac{1}{\gamma(s)} + \frac{\rho^{''}(s)}{\gamma^2(s)\rho(s)}\right]. \tag{62}$$

□

Distribution-aware bound. By using Eq. (55) and approximation of $\log \rho(s+w) - \log \rho(s)$, we have

$$p(w \mid s) \approx \exp\left(aw + \frac{1}{2}bw^2\right) \exp\left(-\frac{1}{2}w^2\gamma(s)\right) \tag{63}$$

$$\propto \exp\left[-\frac{\gamma(s)-b}{2}\left(w - \frac{a}{\gamma(s)-b}\right)^2\right] \tag{64}$$

$$= \mathcal{N}\left(w; \mu_s, \frac{1}{\kappa_s}\right), \tag{65}$$

where $\kappa_s = \gamma(s) - b$ and $\mu_s = \frac{a}{\kappa_s}$. Based on this derivation, the probability of incorrect retrieval satisfies

$$\mathbf{P}[|w| \geq \epsilon \mid s] \approx Q(\sqrt{\kappa_s}\,(\epsilon - \mu_s)) + Q(\sqrt{\kappa_s}\,(\epsilon + \mu_s)) \tag{66}$$

where $Q(\cdot)$ denotes Gaussian Q-function. Consequently,

$$P_{\mathrm{correct}}(s) := \mathbf{P}[|w| < \epsilon \mid s] \tag{67}$$

$$\gtrsim 1 - Q(\sqrt{\kappa_s}\,(\epsilon - \mu_s)) - Q(\sqrt{\kappa_s}\,(\epsilon + \mu_s)) \tag{68}$$

The bound above is substantially tighter, whereas the bound used in the analysis is distribution-free and does not assume Gaussian retrieval errors. Although the bound here is tighter, the distribution-free bound may provide greater intuition.

I.1 Reliable recall

Here we briefly discuss the storage condition of a stimulus. Under the dense memory regime, on average the inter-pattern distance at s is

$$\Delta(s) \sim \frac{1}{\rho(s)}.$$

[1]We say stimulus s is stored whenever $w = |\bar{s} - s|$ is small.

The retrieval noise has standard deviation $\sigma(s) = \frac{1}{\sqrt{\gamma(s)}}$. Define the signal-to-noise ratio:

$$\mathrm{SNR}(s) \coloneqq \frac{\Delta(s)}{\sigma(s)} = \frac{\sqrt{\gamma(s)}}{\rho(s)},$$

which we require $\mathrm{SNR}(s) \gg 1$ for all s to have a reliable recall. One can consider the condition of $\mathrm{SNR}(s) \gg 1$ corresponds to the master inequality in Ramsauer et al. [39][Equation 307].

Remark 1. Prop. 2 characterizes the expected mean squared error *point-wise* in the input stimulus s, rather than as a uniform bound over the stimulus space. This is the key feature distinguishing our result from existing analyses [29, 39], and it has several consequences. With the encoder fixed, the recall rate at a given s is governed by two quantities: an inverse-temperature-like term $\beta(s)$ and a local memory density term $\rho(s)$. Increasing $\beta(s)$ improves recall at s, while recall degrades in regions where few memory patterns provide coverage (small $\rho(s)$). Because both quantities depend on s, the framework directly captures how retrieval success varies across the stimulus space. This point-wise perspective enables analyses that uniform bounds cannot support. In particular, one can study (i) the effect of *non-uniform memory pattern allocation*, where memory density $\rho(s)$ varies across the input space, and (ii) the question of how to optimally *allocate a finite budget* β across stimuli to maximize overall recall. Finally, the term $\beta(s)$ admits a natural biological interpretation as a stimulus-dependent energy budget the animal expends during retrieval. In this sense, our framework extends the modern Hopfield setting toward a more biologically plausible regime, in which both memory coverage and retrieval effort are non-uniform across stimuli [34, 48].

Interpretation. The leading $1/\gamma(s)$ is the irreducible Gaussian retrieval noise. The ρ''/ρ correction is the local curvature of ρ: it lowers MSE near peaks of the memory distribution ($\rho'' < 0$) and raises it near troughs ($\rho'' > 0$). Both bounds tighten as $\gamma(s)$ grows, which is the regime Prop. 1 exploits when minimizing the $\mathcal{L}_2$ loss.

J Extended discussion

Here we discuss extensions of our work that connect to transformers, diffusion models, and computational neuroscience.

Connection to attention modules. Modern transformer attention can be interpreted through the lens of Hopfield retrieval with a fixed inverse temperature [39]. In our framework, a constant β implies that retrieval precision is proportional to encoder Fisher information. Consequently, satisfying the optimal coupling in Thm. 1 requires

$$\mathrm{FI}_{\mathrm{enc}}(s) \propto [\rho^*(s)]^2 \propto p(s)^{\frac{2}{p+2}}. \tag{69}$$

This suggests that constant-temperature attention can achieve the optimal retrieval allocation only when the encoder's representational precision is appropriately matched to memory density. Otherwise, stimulus-dependent inverse temperatures may provide an additional mechanism for achieving this coupling.

Since $\mathrm{FI}_{\mathrm{enc}}(s)$ reflects encoder precision, this yields a testable prediction when applying our framework to a simplified model relevant to modern transformers. We acknowledge that applying the framework to transformers requires a richer analysis to incorporate multidimensional stimuli, see e.g. [55], and we leave this extension to future work. Additionally, modern transformers do not follow clear encoder-decoder structure, and instead have a distributed encoding along the hierarchically stacking attention modules. However, we note as a potential **broader impact** that novel analytical tools for attention modules, even those with simplifications, have the potential to drive more efficient transformer design in future, which may reduce the negative impacts of these technologies.

Characterize biological decoders in cognitive science. The relationship discussed in Sec. 6 in connection with diffusion models leads directly to a testable prediction in cognitive science. Note that the bias-to-variance ratio $b(s)/\mathrm{Var}(\bar{s} \mid s)$ is observable in many psychological experiments. We see here that measuring it allows us to infer the underlying memory density $\rho(s)$ in the decoder. Additionally, under the optimal allocation and by Prop. 1 and Eq. (8), we derive

$$\rho(s) = \frac{M}{\int p(s')^\alpha \, ds'} \, p(s)^\alpha, \tag{70}$$

where α is the optimal allocation exponent from Prop. 1 ($\alpha = 1/(p+2)$ for L^p loss). With some algebraic work, we make the following prediction:

$$\frac{b(s)}{\mathrm{Var}(\bar{s} \mid s)} \approx \alpha \, (\log p)'(s). \tag{71}$$

Because $p(s)$ can also be measured, we propose the proportionality of the bias-variance ratio to the score of the prior as an empirical test for our framework.

Additionally, should this prediction hold, it would provide a novel method for uncovering the (generally unknown) computational objectives of neural populations. This question has been extensively studied through population-coding analyses of biological encoders, e.g., [11, 13, 14, 54]. Notably, Manning et al. [35] analyzed differences across visual brain regions (V1/V2 vs. MT), providing evidence for the hypothesis that computational objectives vary across the visual hierarchy, with early regions prioritizing log losses and later regions shifting toward L^p losses. Despite the broader challenge of identifying the computational goals of neural systems, our framework provides a testable prediction for objective identification in memory recall: by Prop. 1, the value of α in Eq. (71) directly identifies the loss exponent p within the L^p family.

Generalization to high-dimensional stimuli. Our framework has the potential to generalize to high-dimensional stimuli. To facilitate the discussion without a detailed derivation, we provide a sketch of the key steps. Following steps analogous to those in Sec. B, we obtain

$$p(\mathbf{s}_\mu \mid \mathbf{r}) \approx \frac{\exp\left[-\frac{1}{2}(\mathbf{s}_\mu - \hat{\mathbf{s}})^\top \beta(\mathbf{s}) \mathbf{FI}_{\mathrm{enc}}(\mathbf{s})(\mathbf{s}_\mu - \hat{\mathbf{s}})\right]}{\sum_{\mu'} \exp\left[-\frac{1}{2}(\mathbf{s}_{\mu'} - \hat{\mathbf{s}})^\top \beta(\mathbf{s}) \mathbf{FI}_{\mathrm{enc}}(\mathbf{s})(\mathbf{s}_{\mu'} - \hat{\mathbf{s}})\right]}. \tag{72}$$

Under the dense-memory approximation, the partition function in the denominator becomes

$$Z(\hat{\mathbf{s}}) \approx \rho(\hat{\mathbf{s}}) \frac{(2\pi)^{d/2}}{\sqrt{\det[\beta(\mathbf{s})\mathbf{FI}_{\mathrm{enc}}(\mathbf{s})]}}. \tag{73}$$

Here, $\mathbf{FI}_{\mathrm{enc}}(\mathbf{s})$ is the encoder Fisher information matrix. Marginalizing over the asymptotically Gaussian encoder estimate yields the effective retrieval precision matrix

$$\mathbf{\Gamma}(\mathbf{s}) = \frac{\beta(\mathbf{s})}{1+\beta(\mathbf{s})} \mathbf{FI}_{\mathrm{enc}}(\mathbf{s}). \tag{74}$$

Analogous steps can be followed to approximate retrieval objectives under different loss functions. However, the total-resolution constraint must be generalized because retrieval precision is now matrix-valued. One natural extension is a determinant-based constraint,

$$\int \sqrt{\det \mathbf{\Gamma}(\mathbf{s})}\, \mathrm{d}\mathbf{s} = C, \tag{75}$$

which is related to the multidimensional efficient-coding formulation of Yerxa et al. [55].

Although this constraint provides a natural measure of local resolution volume, it does not independently constrain the orientation or anisotropy of the precision matrix. Moreover, under a scalar inverse temperature, the effective retrieval precision matrix remains proportional to the encoder Fisher information matrix, preserving its eigenvectors and relative eigenvalues. A complete multidimensional optimization would therefore require a careful treatment of these geometric constraints, which we leave to future work.

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